SPACE Conferences and Exposition
September 10-12, 2013, San Diego, CA
AIAA SPACE 2013 Conference and Exposition

A Generalized Multi-Commodity Network Flow Model for Space Exploration Logistics

Takuto Ishimatsu*, Olivier de Weck†, and Jeffrey Hoffman‡
Massachusetts Institute of Technology, Cambridge, MA 02139

Yoshiaki Ohkami§
Keio University, Yokohama 223-8526, Japan

Robert Shishko¶
Jet Propulsion Laboratory, California Institute of Technology, Pasadena, CA 91109

Downloaded by UNIVERSITY OF TENNESSEE on October 15, 2015 | http://arc.aiaa.org | DOI: 10.2514/6.2013-5414

In transition to a new era of human space exploration, the question is what the next-generation space logistics paradigm should be. The past studies on space logistics have been mainly focused on a "vehicle" perspective such as propulsive feasibility, cargo capacity constraints, and manifesting strategies, with the arbitrarily predetermined logistics network. But how do we select an optimal logistics network? Especially if we can utilize in-situ resources on the Moon and Mars, it will add complexity to network selection problem. The objective of this paper is to develop a comprehensive graph-theoretic modeling framework to quantitatively evaluate and optimize space exploration logistics from a "network" perspective. In an attempt to create such a modeling framework, we develop a novel network flow model referred to as the generalized multi-commodity network flow (GMCNF) model. On top of the classical network flow problems, the GMCNF model proposed in this paper introduces three types of matrix multiplications (requirement, transformation, and concurrency), and also allows loop edges associated with nodes (graph loops) and multiple edges between the same end nodes (multigraph). With this modification, the model can handle multiple commodities that interact with each other in the form of requirement at nodes, transformation on edges, and concurrency within edges.

A linear programming (LP) formulation of the GMCNF model is applied to human exploration of Mars. First we solve the baseline problem with a demand that is equivalent to that of the NASA's Mars Design Reference Architecture (DRA) 5.0 scenario. It is found that the solution saves 67.5% from the Mars DRA 5.0 reference scenario in terms of the initial mass in low-Earth orbit (IMLEO) primarily because chemical (LOX/LH_2) propulsion is used along with oxygen-rich ISRU. We also present one possible scenario with two "gateway" resource depots at GTO and DTO with orbital transfer vehicles (OTVs) running in the cislunar and Martian systems. Then we solve variant problems that have different settings to see the effect of each factor. Findings include: taking advantage of oxygen-rich ISRU, LOX/LH_2 is preferred to nuclear thermal rocket (NTR), the aerobraking option as well as ISRU availability on the Moon make great contributions in reducing the total mass to be launched from Earth, and as the ISRU production rate decreases, ISRU in each location becomes worthless at a certain threshold and the network topology changes toward direct paths using NTR.

*Postdoctoral Associate, Engineering Systems Division, AIAA Member.
†Professor, Department of Aeronautics and Astronautics & Engineering Systems Division, AIAA Associate Fellow.
‡Professor of the Practice, Department of Aeronautics and Astronautics, AIAA Member.
§Executive Advisor, System Design and Management Research Institute, AIAA Member.
¶Principal Systems Engineer/Economist, Mission Systems Concepts Section, AIAA Member.

Copyright © 2013 by Takuto Ishimatsu. Published by the American Institute of Aeronautics and Astronautics, Inc., with permission.

Nomenclature

$\boldsymbol{A}$ = requirement matrix
$\mathcal{A}$ = set of directed edges
$\boldsymbol{B}$ = transformation matrix
$\boldsymbol{C}$ = concurrency matrix
$\mathcal{G}$ = directed network graph
I_{sp} = specific impulse
$\mathcal{J}$ = objective function
$\mathcal{N}$ = set of nodes
b, $\boldsymbol{b}$ = net supply/demand
c, $\boldsymbol{c}$ = cost per unit flow
c = consumables consumption per crew member per unit time
f_{inert} = inert mass fraction
g_0 = standard gravity ($= 9.80665$ m/s^2)
i = node index ($\in \mathcal{N}$)
j = node index ($\in \mathcal{N}$)
(i, j) = edge from node i to node j ($\in \mathcal{A}$)
k = number of types of commodities
l, $\boldsymbol{l}$ = edge lower bound
l = number of concurrency constraints on each edge
m = number of directed edges
m = mass
n = number of nodes
n = number of variables/constraints
p, q = commodity index ($\in \mathcal{K}$)
r = concurrency constraint index
r_{bo} = boil-off rate
u, $\boldsymbol{u}$ = edge capacity
w = waste generation per crew member per unit time
x, $\boldsymbol{x}$ = flow variable
Δt = duration of edge
ΔV = change in velocity
α = proportional constant for ISRU maintenance mass
β = proportional constant for ISRU resource production
γ = minimum vehicle mass per crew member
η = ratio of structure mass to propellant mass
θ = aeroshell mass fraction
μ = multiplier for generalized network flow
μ = mixture ratio of oxidizer to fuel
ϕ = propellant mass fraction

Superscripts and Subscripts

$(\cdot)^+$ = outflow from tail node
$(\cdot)^-$ = inflow into head node
$(\cdot)^\pm$ = both outflow and inflow
$(\cdot)^{\text{T}}$ = transpose
$(\cdot)_i$ = node i
$(\cdot)_j$ = node j
$(\cdot)_{ii}$ = graph-loop associated with node i
$(\cdot)_{ij}$ = edge (i, j)
$(\cdot)_{pq}$ = (p, q) entry of a matrix
$(\cdot)_{\text{eq}}$ = equality constraint
$(\cdot)_{\text{ineq}}$ = inequality constraint

Downloaded by UNIVERSITY OF TENNESSEE on October 15, 2015 | http://arc.aiaa.org | DOI: 10.2514/6.2013-5414

Downloaded by UNIVERSITY OF TENNESSEE on October 15, 2015 | http://arc.aiaa.org | DOI: 10.2514/6.2013-5414

I. Introduction

WITH the advent of a new era of human space exploration, the research presented in this paper originates from the question of what the next-generation space logistics paradigm should be. Although the Constellation Program has faced cancellation, NASA's new direction for human spaceflight reaffirms that Mars is the ultimate goal of human exploration of the inner solar system. President Obama's space policy speech at Kennedy Space Center in April 2010 discussed the future of US effort in human spaceflight and implied a plan for NASA that follows the Augustine Committee's "Flexible Path to Mars" option first focusing on non-lunar missions, such as near-Earth objects (NEOs), Lagrangian points, and the moons of Mars, followed by exploration of the lunar surface and/or Martian surface.[1] As shown in Figure 1, past human space exploration programs have followed different types of logistics paradigms such as "carry-along" for the Apollo program and "resupply" for the International Space Station (ISS) program. Now, in transition to a new era, the question is, assuming we are ultimately going to Mars, what the next logistics paradigm should be. One possible key to success comes from following a "travel light and live off the land" strategy that has well-served terrestrial explorers for centuries.[2]

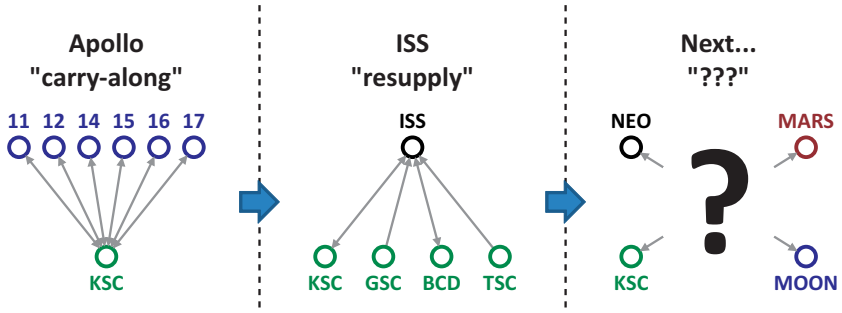


Figure 1. Space logistics paradigms

Adding to the technical challenges of Mars exploration, logistical concerns should be considered far in advance. Far away from home with limited capability to manufacture the basic resources to sustain life, astronauts are constantly at risk of logistical lapses. A well-planned logistics strategy is essential to balance risks, ensure robustness, and achieve maximum exploration capability.[3] The importance of a well-planned logistics strategy is not limited to Mars exploration. Over the next 50 years, mission architectures are expected to transition from single, independent sorties to tightly-integrated campaigns spanning many years and involving several stakeholder organizations. As a system-of-systems, a space exploration campaign will require sophisticated logistics and supply chain planning to maintain human presence in remote, hostile environments.[4] For this reason, space logistics is becoming an important concept for future space exploration.

Compared to terrestrial transportation, space exploration introduces several fundamental differences. First, unlike transports on Earth, space resupply missions are possible only at discrete intervals corresponding to favorable positions of planetary bodies and accommodating lead times required for spacecraft manufacturing, assembly, and ground operations. Second, transports in space can last significantly longer (for weeks to months) compared to terrestrial transports which tend to take no more than hours to days with the exception of ocean freight. Finally, the fraction of usable cargo mass in spaceflight is significantly less than that of terrestrial transports. In summary, to safely explore distant locations in space must respond to a wide range of unexpected events far enough in advance to accommodate launch window opportunities and the long-duration transports while working with severely limited mass capacities. These opposing constraints may require advanced strategies such as completely closed-loop environmental control and life support systems (ECLSS), in-situ resource utilization (ISRU) and manufacturing or repair capabilities, highly efficient packing and container design, significant part commonality between systems, and orbital supply depots.

Although a vast body of research exists for terrestrial transportation networks and supply chain logistics in business and military applications, space logistics is an emerging topic in recent years as we start to

see space exploration not as a set of isolated missions but as an intricately-linked exploration campaign. The literature on space logistics can be divided into the following three categories: (1) terrestrial analogs, (2) mathematical modeling and analysis, and (3) modeling and simulation environment.[5] Terrestrial analog study was mainly focused on the operational aspect such as inventory, radio frequency identification (RFID), extra-vehicular activities (EVA) logistics, and as part of the study classes of supply were formulated to define the categorization of supply items.[6,7] Mathematical modeling and analysis has been carried out in parallel. Theses studies have addressed interplanetary supply chain management, measures of effectiveness (MOEs), and optimal cargo manifesting.[8–12] As a modeling and simulation environment, many efforts have been made to develop a discrete event simulation software called SpaceNet.[13–15]

Of those previous works on space logistics, Taylor et al. solved vehicle design and routing concurrent optimization problem for a two-week lunar sortie in a time-expanded Earth-Moon network with six static nodes using a combinatorial integer programming model.[8] However, it is impractical to apply this modeling technique to a much larger, more complex problem such as Mars exploration with more than 10 static nodes and more than 100 weeks of mission duration. On the other hand, SpaceNet is a software tool that simulates space exploration logistics but in the current version, one needs to pre-define a transportation network and a mission sequence as inputs, and SpaceNet only returns campaign feasibility (propulsive and cargo capacity) and an optimal manifesting strategy.[16]

The past studies mentioned above have been mainly focused on a "vehicle" perspective such as propulsive feasibility, cargo capacity constraints, and manifesting strategies. However, a logistics network was arbitrarily determined without a strong rationale behind the network selection. Then a research question is raised on how we select an optimal logistics network. Especially if we can utilize in-situ resources on the Moon and Mars, it will add complexity to network selection problem rather than a simple round trip between origin and destination. Against this background, we have proposed a graph-theoretic approach to developing a comprehensive modeling framework to quantitatively evaluate and optimize space exploration logistics from a "network" perspective.[5,17] In an attempt to create such a modeling framework, we developed a novel network flow model referred to as the generalized multi-commodity network flow (GMCNF) model,[18] which is presented in this paper. Once the framework for optimization of network selection has been established, it can be implemented as a front end to SpaceNet for logistics network auto-generation.

Figure 2 shows an example of Earth-Moon-Mars logistics network. Surface nodes include Kennedy Space Center (KSC) and Pacific Ocean splashdown (PAC), lunar south pole (LSP), Phobos (PHOB), Deimos (DEIM), and Gale Crater (GC) on Mars, which is the landing site of the JPL's Curiosity, while orbital nodes include low-Earth orbit (LEO), geostationary orbit (GEO) and its transfer orbit (GTO), low lunar orbit (LLO), Earth-Moon Lagrange points (EML), Phobos/Deimos transfer orbits (PTO/DTO), and low Mars orbit (LMO). Edges between those nodes represent possible movements or transports between two locations. Therefore, no direct edge between two nodes means that a direct transport between the two nodes are not allowed. For instance, we must go through LLO to reach the lunar surface.

Then the problem is, given a mission objective, to find the best route(s) in the network. In other words, the optimization result should be able to figure out which nodes and edges to use. This result could be translated to "where to deploy what", providing insights on the best transportation architecture and infrastructure concept. Therefore it should be noted that Figure 2 just includes all the possibilities and it does not necessarily mean we must have something deployed at each node. For this reason, we can add or remove as many nodes and edges as we can think of. One of the goals of this research is to evaluate the potential benefits of in-situ resource utilization (ISRU) from a long-term perspective. Various ISRU systems have been proposed so far such as: hydrogen reduction, methane carbothermal reduction, molten electrolysis (electrowinning), volatile extraction, and polar water extraction on the lunar surface, and Sabatier reaction, reverse water gas shift reaction, and atmosphere electrolysis on the Martian surface.[19–22] The two moons of Mars, Phobos and Deimos, the dwarf planet, Ceres, and near-Earth asteroids could also be sources of raw materials for ISRU.[23,24] As such, ISRU or the ability to produce water, gases, and propellants somewhere halfway to the destination, along with resource depots or the ability to store them, is expected to be one of the most important key concepts in resource-economy in space. Sample images of ISRU plant and space fuel depot are shown in Figure 3. In the graph-theoretic analysis, ISRU capability can be modeled as a mass-origin node where a production rate translates to a mass generation rate.

The graph-theoretic approach essentially models the movement of cargo in a flow network. ISRU allows resources to be available in locations other than Earth surface. Therefore we need to deal separately with

Downloaded by UNIVERSITY OF TENNESSEE on October 15, 2015 | http://arc.aiaa.org | DOI: 10.2514/6.2013-5414

Downloaded by UNIVERSITY OF TENNESSEE on October 15, 2015 | http://arc.aiaa.org | DOI: 10.2514/6.2013-5414

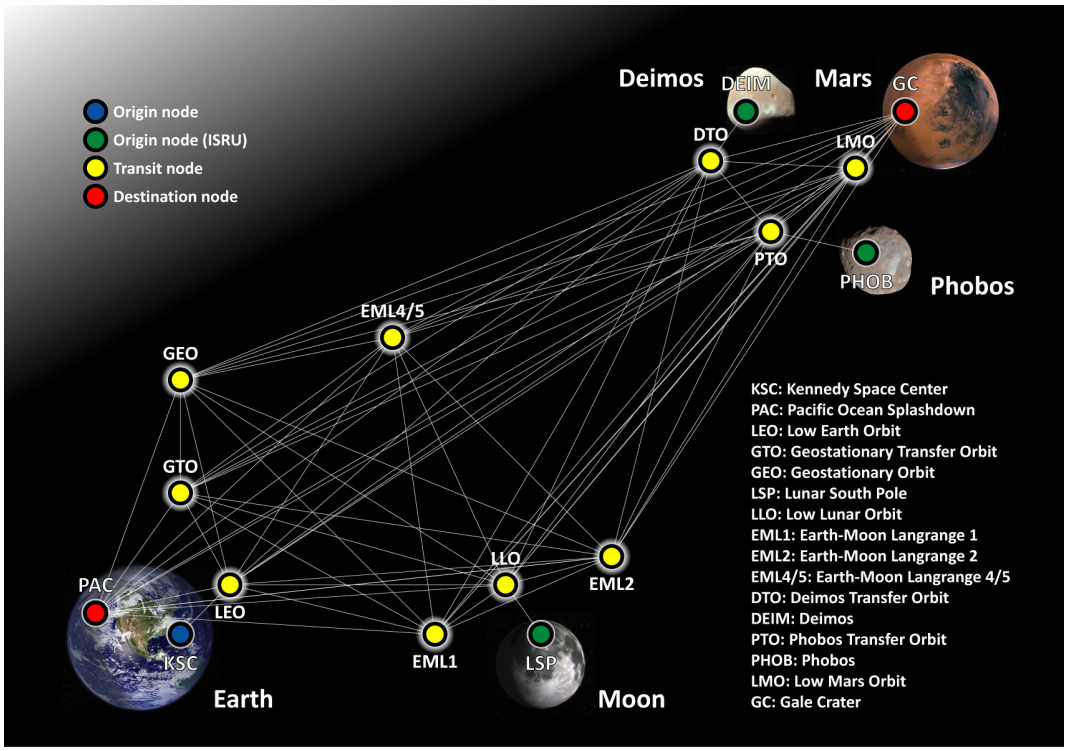


Figure 2. Example of Earth-Moon-Mars logistics network

materials that are only available on Earth (e.g., science payload) and resources that are also available in other locations. For this reason, the problem is formulated as a multi-commodity network flow problem. Section II describes how to formulate a network flow problem and Section III describes how to solve it computationally.

II. Generalized Multi-Commodity Network Flows

This section first reviews classical network flows, followed by a new formulation for generalized multi-commodity network flows (GMCNF). Subsequently a couple of mathematical concepts in graph theory are introduced that help formulating the problem to be addressed in this study.

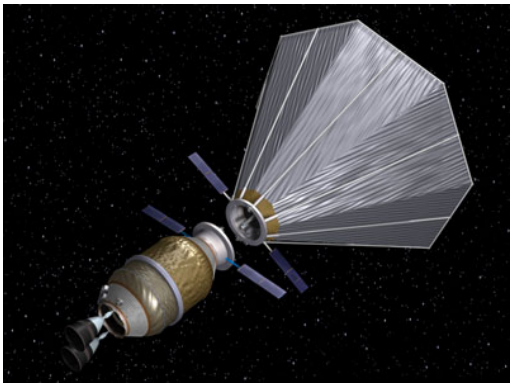
A. Classical Network Flows

While there are many applications of network flow models, the minimum cost flow model is the most fundamental of all network flow problems.[25] The minimum cost flow problem wishes to determine a least cost shipment of a single commodity through a network in order to satisfy demands at certain nodes from available suppliers at other nodes. The special versions of the minimum cost flow problem that play a central role in the theory and applications of network flows include: shortest path problems, maximum flow problems, assignment problems, transportation problems, and circulation problems. Also the generalizations of the minimum cost flow problem include: convex cost flow problems, generalized flow problems, and multi-commodity flow problems.

Let $\mathcal{G} = (\mathcal{N}, \mathcal{A})$ be a directed network defined by a set $\mathcal{N}$ of n nodes and a set $\mathcal{A}$ of m directed edges. Each node $i \in \mathcal{N}$ is associated a number b_i representing its supply/demand. Each edge $(i, j) \in \mathcal{A}$ is associated a *cost* c_{ij} that denotes the cost per unit flow on that edge. If $b_i > 0$, node i is a *supply node*; if $b_i < 0$, node i is a *demand node* with a demand $-b_i$; and if $b_i = 0$, node i is a *transshipment node*. The decision variables in the minimum cost flow problem are edge flows and we represent the flow on an edge $(i, j) \in \mathcal{A}$ by x_{ij}. The minimum cost flow problem is an optimization model formulated as follows:



(a) ISRU plant (image courtesy of NASA JSC)



(b) Space fuel depot (image courtesy of ULA)

Figure 3. Sample images of ISRU plant and space fuel depot

Downloaded by UNIVERSITY OF TENNESSEE on October 15, 2015 | http://arc.aiaa.org | DOI: 10.2514/6.2013-5414

Minimize

$$\mathcal{J} = \sum_{(i,j)\in\mathcal{A}} c_{ij} x_{ij} \tag{1}$$

subject to

$$\sum_{j:(i,j)\in\mathcal{A}} x_{ij} - \sum_{j:(j,i)\in\mathcal{A}} x_{ji} \leq b_i \qquad \forall\ i \in \mathcal{N} \tag{2a}$$

$$l_{ij} \leq x_{ij} \leq u_{ij} \qquad \forall\,(i,j) \in \mathcal{A} \tag{2b}$$

where u_{ij} denotes a *capacity* and l_{ij} denotes a *lower bound*, which is zero in most applications. We refer to the constraints in Eqs. (2a) and (2b) as *mass balance constraints* and *flow bound constraints*, respectively. Prior to introducing the generalized multi-commodity network flow model, two generalizations of the minimum cost flow problem are described below.

1. Generalized Flow Problems

In the minimum cost flow problem described above, we have made one very fundamental, yet almost invisible, assumption: we conserve flow on every edge, that is, the amount of flow on any edge that leaves its tail node equals the amount of flow that arrives at its head node. This assumption is very reasonable in many application settings. Other practical contexts, however, violate this conservation assumption. In generalized flow problems, edges might "consume" or "generate" flow. To address these situations, a positive multiplier μ_{ij} is associated with every edge (i,j) of the network, assuming that if we send x_{ij} unit from node i to node j along the edge (i,j), then $\mu_{ij}x_{ij}$ units arrive at node j. The objective function and constraints remain the same as in the minimum cost flow problem with the exception of the mass balance constraints:

$$\sum_{j:(i,j)\in\mathcal{A}} x_{ij} - \sum_{j:(j,i)\in\mathcal{A}} \mu_{ji} x_{ji} \leq b_i \qquad \forall\ i \in \mathcal{N} \tag{3}$$

If $0 < \mu_{ij} < 1$, the edge is *lossy* and if $1 < \mu_{ij} < \infty$, the edge is *gainy*. If every multiplier has value 1, the generalized network flow model becomes the minimum cost flow problem.

2. Multi-Commodity Flow Problems

Up to this point, network models composed of a single commodity have been considered. In many application contexts, however, several physical commodities, vehicles, or messages, each governed by their own network flow constraints, share the same network. Using the vector notations (bold letters) instead of scalar expression, Eqs. (1), (2a), and (2b) can be rewritten as:
Minimize

$$\mathcal{J} = \sum_{(i,j)\in\mathcal{A}} \boldsymbol{c}_{ij}{}^{\mathrm{T}} \boldsymbol{x}_{ij} \tag{4}$$

subject to

$$\sum_{j:(i,j)\in\mathcal{A}} \boldsymbol{x}_{ij} - \sum_{j:(j,i)\in\mathcal{A}} \boldsymbol{x}_{ji} \leq \boldsymbol{b}_i \qquad \forall\, i \in \mathcal{N} \tag{5a}$$

$$\boldsymbol{l}_{ij} \leq \boldsymbol{x}_{ij} \leq \boldsymbol{u}_{ij} \qquad \forall\, (i,j) \in \mathcal{A} \tag{5b}$$

If k is the number of types of commodities to be considered, each vector is a k-dimensional column vector. However, if the commodities do not interact in any way, then this is identical to solving each single-commodity problem separately. In some situations, the individual commodities share the common edge capacities, that is, each edge has a capacity that restricts the total flow of all commodities on that edge while in other applications including the one described later in Section IV, the individual commodities interact in other ways. The next section describes a new network flow model that allows the interaction between commodities.

B. Generalized Multi-Commodity Network Flows (GMCNF)

The generalized multi-commodity network flow (GMCNF) problem is literally a mix of the generalized flow problem and the multi-commodity flow problem in the previous section. As discussed earlier, the problem addressed in this study encompasses the characteristics of the two flows and therefore should be formulated as a generalized multi-commodity network flow problem because (1) major commodities such as propellant are substantially consumed during transport (generalized flows), and (2) a space exploration mission carries along multiple commodities, which could even have different origin-destination pairs especially when considering ISRU (multi-commodity flows).

One very important addition in the generalized multi-commodity network flow is the interaction between different commodities. In generalized flow problems, a single commodity could increase or decrease its quantity by passing through an edge. In generalized multi-commodity flow problems, however, we must consider not only a gain or loss of each commodity itself but also a gain or loss of a commodity due to another commodity, and even a transformation between different commodities. For example, the amount of propellant consumed is driven by the total mass (not only the propellant itself), and food is consumed and turned into waste by the crew. This can all be mathematically implemented by multiplying a flow vector $\boldsymbol{x}_{ij}$ by a square matrix instead of a scalar μ_{ij} as in Eq. (3). The off-diagonal entries of this matrix should be able to capture the interactions between different commodities.

Before formulating the problem, let us introduce two more additions to generalize the model more completely. When sending out a flow $\boldsymbol{x}_{ij}$ from node i, we might have additional requirements other than $\boldsymbol{x}_{ij}$ itself. The other addition is a flow concurrency constraint. When commodity 1 flows on the edge, a certain amount of commodity 2 might also have to be carried along with commodity 1. For example, in order to perform aerobraking maneuver at Mars arrival, a spacecraft would need an aeroshell whose mass should be around 15% of the total mass being braked.



Figure 4. Outflow $\boldsymbol{x}_{ij}^{+}$, inflow $\boldsymbol{x}_{ij}^{-}$, and unit costs for outflow $\boldsymbol{c}_{ij}^{+}$ and inflow $\boldsymbol{c}_{ij}^{+}$

Figure 4 shows the building blocks of the generalized multi-commodity network flows. In order to explicitly incorporate the above three additions (transformation, requirement, and concurrency), we split the flow into two parts: $\boldsymbol{x}_{ij}^{+}$ and $\boldsymbol{x}_{ij}^{-}$, where $\boldsymbol{x}_{ij}^{+}$ represents the outflow from node i and $\boldsymbol{x}_{ij}^{-}$ represents the inflow into node j. $\boldsymbol{c}_{ij}^{+}$ and $\boldsymbol{c}_{ij}^{+}$ denote the unit costs for outflow and inflow, respectively. Using this notation, the generalized multi-commodity network flow can be formulated as follows:
Minimize

$$\mathcal{J} = \sum_{(i,j)\in\mathcal{A}} \left({\boldsymbol{c}_{ij}^{+}}^{\mathrm{T}} \boldsymbol{x}_{ij}^{+} + {\boldsymbol{c}_{ij}^{-}}^{\mathrm{T}} \boldsymbol{x}_{ij}^{-} \right) \tag{6}$$

subject to

$$\sum_{j:(i,j)\in\mathcal{A}} \boldsymbol{A}_{ij}^{+} \boldsymbol{x}_{ij}^{+} - \sum_{j:(j,i)\in\mathcal{A}} \boldsymbol{A}_{ji}^{-} \boldsymbol{x}_{ji}^{-} \leq \boldsymbol{b}_i \qquad \forall\, i \in \mathcal{N} \tag{7a}$$

Downloaded by UNIVERSITY OF TENNESSEE on October 15, 2015 | http://arc.aiaa.org | DOI: 10.2514/6.2013-5414

$$\boldsymbol{B}_{ij}\boldsymbol{x}_{ij}^{+} = \boldsymbol{x}_{ij}^{-} \qquad \forall\,(i,j) \in \mathcal{A} \tag{7b}$$

$$\boldsymbol{C}_{ij}^{+}\boldsymbol{x}_{ij}^{+} \leq \boldsymbol{0} \quad \text{and} \quad \boldsymbol{C}_{ij}^{-}\boldsymbol{x}_{ij}^{-} \leq \boldsymbol{0} \qquad \forall\,(i,j) \in \mathcal{A} \tag{7c}$$

$$\boldsymbol{0} \leq \boldsymbol{x}_{ij}^{+} \leq \boldsymbol{u}_{ij}^{+} \quad \text{and} \quad \boldsymbol{0} \leq \boldsymbol{x}_{ij}^{-} \leq \boldsymbol{u}_{ij}^{-} \qquad \forall\,(i,j) \in \mathcal{A} \tag{7d}$$

We introduce three matrices: $\boldsymbol{A}_{ij}^{\pm}$, $\boldsymbol{B}_{ij}$, and $\boldsymbol{C}_{ij}^{\pm}$. We call $\boldsymbol{A}_{ij}^{\pm}$ a *requirement matrix*, $\boldsymbol{B}_{ij}$ a *transformation matrix*, and $\boldsymbol{C}_{ij}^{\pm}$ a *concurrency matrix*. As can be seen from the constraints in Eqs. (7a) and (7b), $\boldsymbol{A}_{ij}^{+}\boldsymbol{x}_{ij}^{+}$ is required at node i to send outflow $\boldsymbol{x}_{ij}^{+}$ into edge (i,j), $\boldsymbol{x}_{ij}^{+}$ is transformed into $\boldsymbol{x}_{ij}^{-} = \boldsymbol{B}_{ij}\boldsymbol{x}_{ij}^{+}$, and $\boldsymbol{A}_{ij}^{-}\boldsymbol{x}_{ij}^{-}$ is received at node j with inflow $\boldsymbol{x}_{ij}^{-}$ from edge (i,j). Also in the concurrency constraints in Eq. (7c), the relationship between commodities on each edge (i,j) is self-constrained such that the dot product with $\boldsymbol{C}_{ij}^{\pm}$ is less than or equal to zero. With this modification, we can treat the flow gain and loss due to the interaction between commodities, transformation between commodities, additional requirements at nodes, and flow concurrency on edges. If we consider k types of commodities, that is, the decision variable vector includes k components, then the $\boldsymbol{A}_{ij}^{\pm}$ and $\boldsymbol{B}_{ij}$ matrices must be k-by-k square matrices while the $\boldsymbol{C}_{ij}^{\pm}$ matrix is a l-by-k matrix, where l is the number of concurrency constraints on edge (i,j).

Note that if $\boldsymbol{A}_{ij}^{\pm}$ and $\boldsymbol{B}_{ij}$ are identity matrices and $\boldsymbol{C}_{ij}^{\pm}$ is a null matrix, the constraints in Eqs. (7a)-(7d) turn back to the constraints in the classical multi-commodity flow model in Eqs. (5a)-(5b). Also, if $\boldsymbol{A}_{ij}^{\pm}$ and $\boldsymbol{B}_{ij}$ are diagonal matrices and $\boldsymbol{C}_{ij}^{\pm}$ is a null matrix, there are no interactions between commodities and therefore the problem can be decoupled into k independent generalized flow problems. To put it the other way around, the off-diagonal entries of $\boldsymbol{A}_{ij}^{\pm}$ and $\boldsymbol{B}_{ij}$ and non-zero entries of $\boldsymbol{C}_{ij}^{\pm}$ indicate that there are interactions between commodities. For example, the (p,q) entry of $\boldsymbol{A}_{ij}^{+}$ means that if we send 1 unit of the qth commodity, we also need $[A_{pq}]_{ij}^{+}$ units of the pth commodity. The (r,p) entry of $\boldsymbol{C}_{ij}^{+}$, which is $[C_{rp}]_{ij}^{+}$, is the pth commodity's contribution to the rth concurrency constraint.

1. *Requirement Matrix* $\boldsymbol{A}_{ij}^{\pm}$

The $\boldsymbol{A}_{ij}^{\pm}$ matrix can describe the commodities that are additionally required by sending out or receiving the flow but do not flow on the edge themselves. Examples of this include the electricity consumed for pumping freshwater into the pipeline, the money received from the sales of commercial products, and the workforce required for loading/unloading a freighter at the seaport. Another example is a "plant" such as an ISRU plant. In this study, a generic plant as a resource processing facility is modeled using a "graph-loop" instead of a node. As discussed later in Section IV, the $\boldsymbol{A}$ matrix also plays a key role in a graph-loop.

2. *Transformation Matrix* $\boldsymbol{B}_{ij}$

The $\boldsymbol{B}_{ij}$ matrix can describe the flow gain/loss or transformation between commodities. Unless the $\boldsymbol{B}_{ij}$ matrices are all identity matrices, there is no longer the underlying law of flow conservation. Examples of this include propulsive burn, resource boil-off, crew consumables consumption and waste generation, ISRU resource production, and so on. When there are multiple flow transformation events on a single edge, $\boldsymbol{B}_{ij}$ can be the product of multiple matrices, in which the subsequent transformation matrix is multiplied from the left:

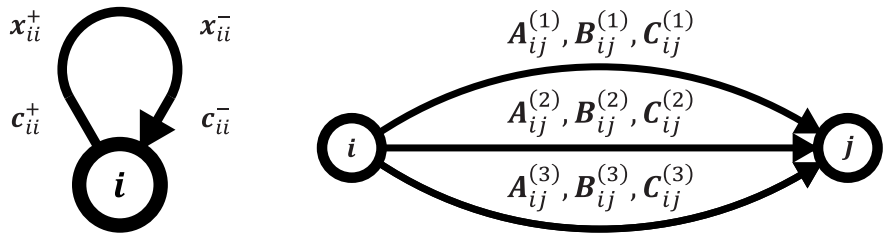
$$\boldsymbol{B}_{ij} = \boldsymbol{B}_{ij}^{(n)} \cdots \boldsymbol{B}_{ij}^{(2)} \boldsymbol{B}_{ij}^{(1)} \tag{8}$$

In general, a non-diagonal matrix is not commutative. Therefore, if there are interactions between different commodities, that is, $\boldsymbol{B}_{ij}$ matrices have off-diagonal entries, then the order of matrix multiplication must be exactly the sequence of transformation events.

3. *Concurrency Matrix* $\boldsymbol{C}_{ij}^{\pm}$

When a commodity flows on an edge, it might need a certain amount of another commodity to travel along with it. As described later in Section IV, the $\boldsymbol{C}_{ij}^{\pm}$ matrix is used to enforce mass constraints on crew transfer vehicle, structure for propellant, and aeroshell.

Downloaded by UNIVERSITY OF TENNESSEE on October 15, 2015 | http://arc.aiaa.org | DOI: 10.2514/6.2013-5414



(a) A graph loop (b) Multiple edges (parallel edges) between nodes i and j

Figure 5. Other concepts in graph theory

C. Other Concepts in Graph Theory

A couple of mathematical concepts in graph theory that help formulating the problem to be addressed in this study are also introduced in this section.

1. Graph Loop

In graph theory, a graph loop is an edge that connects a node to itself (Figure 5(a)). It is also called a self-loop or a "buckle". This can be used for modeling a generic plant as a resource processing facility. A resource processing facility is likely to be modeled as a node, as opposed to a typical edge modeling transportation. In this study, however, a resource processing facility is also modeled as an edge connecting a node to itself. $\boldsymbol{ABC}$ matrices that were discussed in the previous section are also applicable to graph loops and therefore resource processing can be represented by a requirement matrix $\boldsymbol{A}$ and a transformation matrix $\boldsymbol{B}$. Examples of a graph loop used in the space application (discussed later in Chapter IV) include (1) ISRU on the surface nodes producing resource and requiring maintenance and (2) Mars surface exploration consuming oxygen, water, and food, and generating waste.

2. Multigraph

In graph theory, an undirected graph that has no loops and no more than one edge between any two different nodes is called a simple graph. As opposed to a simple graph, a multigraph refers to a graph in which multiple edges (also called "parallel edges") are either permitted or required between the same end nodes. Thus, as shown in Figure 5(b), two nodes may be connected by more than one edge. For example, if we think about an interplanetary transfer from LEO to LMO, we have multiple choices on propulsion system such as chemical, nuclear thermal rocket (NTR), solar electric propulsion (SEP), and so on, each of which has different I_{sp}. We also have aerocapture option; if we perform aerocapture, we need to bring an appropriate aeroshell. Moreover, there are trade-offs between ΔV and TOF (time of flight); if we send a robotic mission, we might prefer a lower ΔV with a longer TOF while if we send human, we might prefer a shorter TOF with a compromise of ΔV. All these parameters are implemented through $\boldsymbol{ABC}$ matrices on each edge. Therefore such multiple choices can be modeled by allowing parallel edges between the same end nodes. By implementing parallel edges that represent trades, the optimization of network flow will automatically give us a solution to the trade problem.

III. Formulation of Optimization Problem

In the previous section, we presented the generalized multi-commodity network flow model using a requirement matrix $\boldsymbol{A}_{ij}^{\pm}$, a transformation matrix $\boldsymbol{B}_{ij}$, and a concurrency matrix $\boldsymbol{C}_{ij}^{\pm}$ on a multigraph that allows loops and multiple edges between the same two nodes. This section discusses how to actually implement and solve the optimization problem based on those mathematical formulations. While there are many fields of optimization, in order to take advantage of linearity of the constraints in Eqs. (7), this study formulates and solves linear programming (LP) problems.

Downloaded by UNIVERSITY OF TENNESSEE on October 15, 2015 | http://arc.aiaa.org | DOI: 10.2514/6.2013-5414

Downloaded by UNIVERSITY OF TENNESSEE on October 15, 2015 | http://arc.aiaa.org | DOI: 10.2514/6.2013-5414

A. LP Formulation for GMCNF

Linear programming (LP) is an optimization problem with a linear objective function, a set of linear equality and inequality constraints, and a set of nonnegativity restrictions imposed upon the underlying decision variables. Its feasible region is a convex polyhedron, which is a set defined as the intersection of finitely many half spaces, each of which is defined by a linear inequality. LP algorithm finds a corner point in the polyhedron where the objective function has the smallest (or largest) value if such a point exists.

Linear programs are problems that can be expressed in canonical form:
Minimize

$$\mathcal{J} = \boldsymbol{c}^{\mathrm{T}}\boldsymbol{x} \tag{9}$$

subject to

$$\boldsymbol{A}_{\mathrm{ineq}}\boldsymbol{x} \leq \boldsymbol{b}_{\mathrm{ineq}} \tag{10a}$$

$$\boldsymbol{A}_{\mathrm{eq}}\boldsymbol{x} = \boldsymbol{b}_{\mathrm{eq}} \tag{10b}$$

$$\boldsymbol{x} \geq \boldsymbol{0} \tag{10c}$$

where $\boldsymbol{x}$ represents the vector of variables (to be determined), $\boldsymbol{c}$, $\boldsymbol{b}_{\mathrm{ineq}}$, and $\boldsymbol{b}_{\mathrm{eq}}$ are vectors of coefficients (known), and $\boldsymbol{A}_{\mathrm{ineq}}$ and $\boldsymbol{A}_{\mathrm{eq}}$ are matrices of coefficients (known). Eq. (10a) represents inequality constraints, Eq. (10b) equality constraints, and Eq. (10c) nonnegativity constraints. If this problem has n_x nonnegative decision variables, n_{ineq} inequality constraints, and n_{eq} equality constraints, then the vectors $\boldsymbol{c}$ and $\boldsymbol{x}$ are n_x-dimensional column vectors, the matrix $\boldsymbol{A}_{\mathrm{ineq}}$ has n_{ineq} rows and n_x columns, the matrix $\boldsymbol{A}_{\mathrm{eq}}$ has n_{eq} rows and n_x columns, and the vectors $\boldsymbol{b}_{\mathrm{ineq}}$ and $\boldsymbol{b}_{\mathrm{eq}}$ are n_{ineq}-dimensional and n_{eq}-dimensional column vectors, respectively.

Generalized multi-commodity network flow problems defined in Eqs. (6) and (7) can be rewritten in canonical form by translating the constraints in Eqs. (7) into equality and inequality constraints above. The transformation constraints in Eq. (7b) fall into equalities while the rest of the constraints fall into inequalities. Figure 6 summarizes this translation. The cells shaded in light red configure the $\boldsymbol{A}_{\mathrm{ineq}}$ while the cells shaded in light yellow configure the $\boldsymbol{A}_{\mathrm{eq}}$ matrix.

Variables	$\boldsymbol{x}_{ij}^{+}$	$\boldsymbol{x}_{ij}^{-}$	$\boldsymbol{x}_{hi}^{-}$		RHS	Σ	$\forall$
Objective Function	$\boldsymbol{c}_{ij}^{+}$	$\boldsymbol{c}_{ij}^{-}$	$\boldsymbol{0}$	$=$	$\mathcal{J}$	$(i,j) \in \mathcal{A}$	
Mass Balance (Net Supply/Demand)	$\boldsymbol{A}_{ij}^{+}$	$\boldsymbol{O}$	$-\boldsymbol{A}_{hi}^{-}$	$\leq$	$\boldsymbol{b}_i$	$j:(i,j) \in \mathcal{A}$ $h:(h,i) \in \mathcal{A}$	$i \in \mathcal{N}$
Self-Constraints for Outflow	$\boldsymbol{C}_{ij}^{+}$	$\boldsymbol{O}$	$\boldsymbol{O}$	$\leq$	$\boldsymbol{0}$		$(i,j) \in \mathcal{A}$
Self-Constraints for Inflow	$\boldsymbol{O}$	$\boldsymbol{C}_{ij}^{-}$	$\boldsymbol{O}$	$\leq$	$\boldsymbol{0}$		$(i,j) \in \mathcal{A}$
Flow Transformation	$\boldsymbol{B}_{ij}$	$-\boldsymbol{I}$	$\boldsymbol{O}$	$=$	$\boldsymbol{0}$		$(i,j) \in \mathcal{A}$

Figure 6. Generalized multi-commodity network flow LP

B. Computational Complexity

Suppose a generalized multi-commodity network flow problem has n nodes, m edges, and k types of commodities. Since the flow is split into outflow and inflow for each edge, there are $2m$ flow vectors and $2m$ cost vectors, each of which is a k-dimensional column vector. In order to formulate an LP problem in canonical form, we need to put them together to construct one large column vector so that we can compute the objective function in Eq. (9) by one large vector multiplication, that is, $\boldsymbol{x}_{ij}^{+}$ and $\boldsymbol{x}_{ij}^{-}$ build up a $2km$-dimensional column vector $\boldsymbol{x}$, and $\boldsymbol{c}_{ij}^{+}$ and $\boldsymbol{c}_{ij}^{-}$ build up a $2km$-dimensional column vector $\boldsymbol{c}$. This is equivalent to an LP problem with $2km$ scalar variables.

As discussed earlier in Section II, the $\boldsymbol{A}_{ij}^{\pm}$ and the $\boldsymbol{B}_{ij}$ matrices are k-by-k square matrices while the $\boldsymbol{C}_{ij}^{\pm}$ matrix is a l-by-k matrix, where l is the number of concurrency constraints on edge (i, j). In actuality, the number of concurrency constraints could be different with each edge. However, for simplicity, let l denote the maximum number of concurrency constraints among edges, and for edges with less than l concurrency constraints, we can just assign zeros to avoid the unnecessary rows without any influence on the solution. Therefore there are at most lm concurrency constraints for outflows and inflows, respectively. Since the mass balance constraints are defined for each node, we have kn mass balance constraints. Putting them together, the matrix $\boldsymbol{A}_{\text{ineq}}$ has $(kn + 2lm)$ rows and $2km$ columns. On the other hand, since the flow transformation is defined for each edge, there are km transformation constraints in total. Hence, the matrix $\boldsymbol{A}_{\text{eq}}$ has km rows and $2km$ columns. To summarize, the generalized multi-commodity network flow problem has $2km$ variables, $(kn + 2lm)$ inequality constraints, and km equality constraints as well as $2km$ flow bound constraints. Thus, the problem size is determined by the four parameters: the number of nodes n, the number of edges m, the number of types of commodities to be considered k, and the maximum number of concurrency constraints among edges l. For example, a fully connected multigraph with 10 nodes has 100 edges including 10 loops. If we consider 5 types of commodities and if we assume 1 concurrency constraint for each outflow and inflow, this problem has 1000 variables, 250 inequality constraints, and 500 equality constraints in addition to 1000 flow bound constraints.

While the computational time highly depends on the method that the solver uses, the computational complexity is apparently a linear function of the number of nodes n, the number of edges m, and the number of commodities k because the number of variables can be represented by $2km$ and the number of both inequality and equality constraints can be represented by $kn + km + 2lm$. The number of commodities k and the number of edges m seem to have a relatively big impact on the computational complexity compared to the number of nodes n. However, since the number of nodes and the number of edges are closely related, n can greatly contribute to the problem size through both m and itself, depending on the network topology.

C. Implementation

In this study, the GMCNF model is implemented in MATLAB®. Whereas various numerical algorithms are available within the MATLAB Optimization Toolbox™, this study employs the IBM® ILOG® CPLEX® Optimization Studio using the MATLAB connector.

First, a dataset including supply/demand, parameters, and assumptions is created using Microsoft Office Excel®. Then the dataset is fed into MATLAB and a generalized multi-commodity network described in Section II is constructed by defining supplies/demands at nodes, objective function coefficients, capacities, and the $\boldsymbol{ABC}$ matrices for edges. The GMCNF problem is translated into an LP as described in Section A. The CPLEX function to solve LP problems is `cplexlp`, using the following syntax:

```
x = cplexlp(f, Aineq, bineq, Aeq, beq, lb, ub)
```

where `f` is equivalent to $\boldsymbol{c}$ in Eq. (9), `Aieq` and `Aeq` are built up from the cells in light red and light yellow in Table 6, respectively, `bineq` and `beq` are their corresponding RHSs, `lb` is the lower bound, and `ub` is the edge capacity.

IV. Case Study: Mars Exploration Logistics

Since all the mathematical preparation has been presented up to this point, this section presents the application of the GMCNF LP model to the logistics problem for human exploration of Mars. One advantage of the LP formulation is that it can be quickly solved and that if solved, the solution is guaranteed to be optimal. This advantage of LP allows us to solve a number of instances.

A. Background

Since the general introduction about space exploration logistics has been presented in Section I, this section discusses a more specific problem description. As stated in Section I, the objective of this study is to develop a comprehensive graph-theoretic modeling framework to quantitatively evaluate and optimize space exploration logistics from a network perspective. In the past studies a logistics network seems to have been arbitrarily determined without a strong rationale behind it while this study considers all the possibilities

Downloaded by UNIVERSITY OF TENNESSEE on October 15, 2015 | http://arc.aiaa.org | DOI: 10.2514/6.2013-5414

in the logistics network selection. Figure 2 in Section I shows the Earth-Moon-Mars logistics network with several representative nodes. Strictly speaking, we can define an infinite number of orbital nodes because, for example, we can think of an Earth orbit with any altitude. However, if we include an unnecessarily large number of nodes in the problem, which then creates a huge number of edges as well, it makes the problem size very large as discussed in Section III. For this reason, it is wise to limit the number of nodes and edges to be considered and select a set of reasonably representative nodes. This case study discusses the space logistics network for human exploration of Mars, based on a set of nodes in Figure 2.

1. Mars Design Reference Architecture 5.0

NASA has undertaken substantial effort to develop a design reference architecture (DRA) for conceptual missions to Mars.[26] DRA represents the current "best" strategy for human missions and architectures and should be constantly updated as we learn. It also serves as a benchmark against which alternative architectures can be measured. As of mid-2013, the most recent publication is Mars DRA 5.0, which was published in July 2009.[27] This design reference describes the spacecraft and missions which could be used for the first three excursions to the surface of Mars. The Mars exploration architecture is heavily based on lunar concepts from the Constellation Program, including the Ares V heavy lift launch vehicle, but also includes advanced technology concepts such as nuclear thermal rockets (NTR) for interplanetary propulsion, zero-loss cryogenic coolers for propellant transportation, aerocapture as the Mars arrival capture method, in-situ resource utilization (ISRU) for Mars ascent propellant production, and nuclear fission reactors for surface power.

2. Cislunar Propellant and Logistics Infrastructure

On the other hand, various concepts for a cislunar propellant and logistics infrastructure and transportation architecture have recently been proposed. Such concepts include a propellant depot, a reusable lunar lander (RLL), a propellant tanker, an orbital transfer vehicle (OTV) with aerobraking capability.[28,29] These concepts are "game-changing" because they would fundamentally affect the architecture of future space operations, providing greater access to space beyond Earth orbit. The challenge is to create the infrastructure with minimum development costs and to assure that operational costs do not diminish its benefits. These cislunar infrastructure/architecture concepts are never used in Mars DRA 5.0. Therefore one of the goals of this case study is to discuss the potential benefits of utilizing lunar resources and cislunar network for human Mars missions. As shown in Figure 2, there is also the Martian system including two moons. While there are various views about the possibilities of ISRU on Phobos and Deimos, we assume that surface regolith and/or water ice in the interior are available on these two moons so that we can evaluate the potential utility of Phobos/Deimos. If the cislunar system would be useful for outbound missions to Mars, it would not be surprising that the Martian system is useful for return trip to Earth.

B. Network Graph, Decision Variables, and Objective Functions

Figure 7 shows a notional network graph that extracts only the relationship between nodes and edges from a logistics network in Figure 2. This network graph includes 16 nodes in total. This network graph allows both multiple edges between the same end nodes and graph-loops associated with nodes. For example, we have multiple choices on propulsion system with aerocapture option for each edge. We consider two propulsion systems: conventional chemical propulsion (LOX/LH_2) and nuclear thermal rocket (NTR). One of the advantages of the GMCNF model is that system-level trades such as the selection of propulsion system are automatically conducted in the network flow optimization by implementing multiple choices as multiple edges. Of course these multiple choices have different characteristics and constraints that must be individually translated into the $\boldsymbol{ABC}$ matrices. In total, there are 357 directed edges including 9 loops. However, note that not necessarily all nodes and edges will be used in the optimal solution.

First, decision variables included in the flow vector and an objective function need to be selected. In the GMCNF model, everything that moves around the network (even the crew) is referred to as a "commodity".

Downloaded by UNIVERSITY OF TENNESSEE on October 15, 2015 | http://arc.aiaa.org | DOI: 10.2514/6.2013-5414

Downloaded by UNIVERSITY OF TENNESSEE on October 15, 2015 | http://arc.aiaa.org | DOI: 10.2514/6.2013-5414

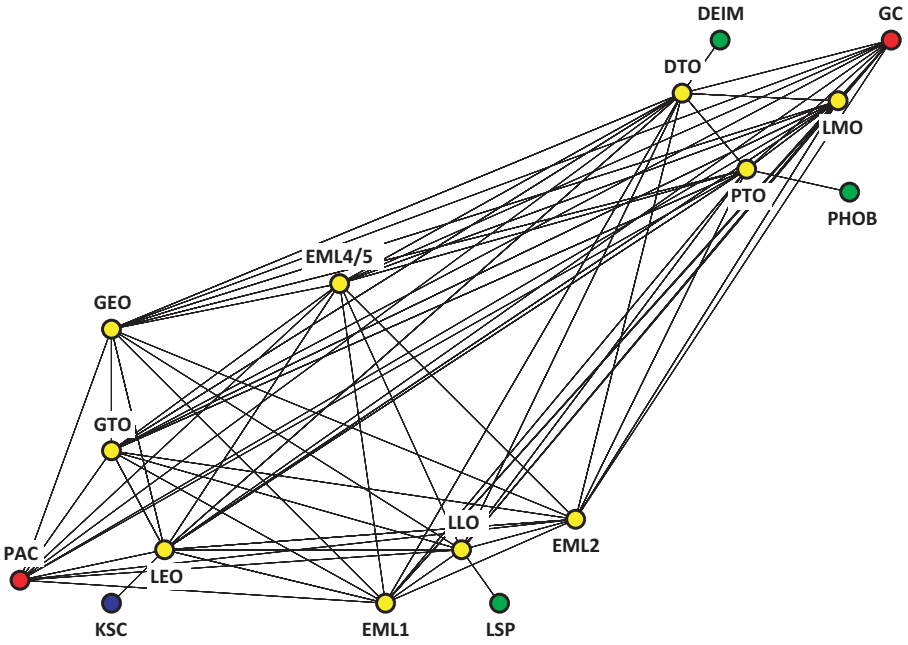


Figure 7. Network graph with 16 nodes and 357 edges (including 9 loops)

The decision variables can be divided into the following categories:

$$\boldsymbol{x}_{ij}^{\pm} = \begin{bmatrix} \text{crew} \\ \text{transportation} \\ \text{resources} \\ \text{infrastructure} \end{bmatrix}_{ij}^{\pm} \tag{11}$$

Each category has sub-classes as follows:

$$\text{crew} = \begin{bmatrix} \text{crew} \\ \text{crewRe} \end{bmatrix} \tag{12a}$$

$$\text{transportation} = \begin{bmatrix} \text{vehicle} \\ \text{habitat} \\ \text{inertLOX} \\ \text{inertNTR} \\ \text{aeroshell} \end{bmatrix} \tag{12b}$$

$$\text{resources} = \begin{bmatrix} \text{hydrogen} \\ \text{oxygen} \\ \text{water} \\ \text{food} \\ \text{waste} \end{bmatrix} \tag{12c}$$

$$\text{infrastructure} = \begin{bmatrix} \text{plant} \\ \text{maintenance} \end{bmatrix} \tag{12d}$$

where "crewRe" represents crew returning to Earth distinguished from crew traveling outbound (the reason for this is described later), "transportation" includes a crew transfer vehicle, a surface habitat, an inert mass for LOX/LH_2 and NTR (separately defined), and an aeroshell system, "resources" includes hydrogen, oxygen, water as well as crew provisions and waste, and "infrastructure" represents an ISRU plant and maintenance mass. These variables are all evaluated in mass [kg].

While we could define any objective function with respect to which the network flow is optimized, we simply use the total mass to be launched from Earth, which in this case study is Kennedy Space Center (KSC). Using the notation of the GMCNF model in Eq. (6), this can be written as:

$$\boldsymbol{c}_{ij}^{+} = \begin{cases} \mathbf{1} & \text{for } i = \text{KSC} \\ \mathbf{0} & \text{otherwise} \end{cases} \qquad \text{and} \qquad \boldsymbol{c}_{ij}^{-} = \mathbf{0} \quad \forall\,(i,j) \in \mathcal{A} \tag{13}$$

where **1** and **0** are vectors. Since KSC is only connected to LEO, this is essentially equivalent to initial mass in LEO (IMLEO), which is often used as a measure of the mission cost as a widely accepted surrogate for estimating it.

C. Supply and Demand

Next we define supply/demand $\boldsymbol{b}_i$ at each node. In the network in Figure 7, Kennedy Space Center (KSC), the lunar south pole (LSP), Phobos (PHOB), Deimos (DEIM), and Gale Crater (GC) on Mars are the supply nodes (sources). While KSC can provide all the commodities in Eqs. (12), the other nodes can only provide resources such as hydrogen, oxygen, and water. Surface manufacturing with in-situ resources has also been proposed but this case study only considers resource production.[19] We assume that KSC has an infinite supply of all the commodities except for "crewRe". For the ISRU nodes, raw materials are infinitely available but the amount of resources produced depends on the capacity of the ISRU system. For this reason, we must enforce some constraint on this instead of setting the supply of resources to infinity. As explained earlier, a facility is modeled as a loop edge. ISRU can be modeled in this way: when a plant of a mass of x is sent into a loop edge, it returns the flow that includes a resource of a mass of βx, where β is a proportional constant. There is an assumption behind this that the ISRU plant is linearly scalable. This is why we explicitly include "plant" in the decision variables. If we utilize the ISRU system somewhere, we can produce resources depending on the size of the plant but at the same time we must continuously supply the maintenance to the plant, which is also explicitly included in the variables.

Defining the demand itself is essentially identical to defining the mission. If we have a demand of "plant" at LSP, it is a lunar mission to send a plant on the lunar south pole; if we have a demand of "crew" on Mars, it is a manned mission to Mars. In this study we choose human exploration of Mars as a case study and the demand is determined by reference to Mars DRA 5.0.[27] While the mission objectives of Mars DRA 5.0 range from science such as search for life, current and ancient climate, geology and geophysics to engineering such as long-duration human spaceflight and sustained human presence on Mars, the mission from a logistics perspective is to send crew of six to Martian surface and to bring them back to Earth after about 500 days of stay on Mars. We have the demand of "crewRe" at PAC instead of having the demand of "crew" at GC (if we set the demand of "crew" at GC, it is interpreted as a one-way mission or the outbound portion of a decoupled problem). Therefore we do not have the demand of "crew" on Mars, but we must also deploy the surface habitat (SHAB) that supports the crew for the long stay on the surface of Mars. Unlike the crew, the SHAB would not be brought back to Earth or would even be useful for the subsequent human presence on Mars. Hence, we translate it into the demand of "habitat" at GC. The total mass of the surface systems amounts to 51,700 kg and we use this value in a lump as a demand of "habitat" on the surface of Mars. There is no need to define other demands such as crew consumables or ISRU maintenance mass because these commodities are required or "consumed" through the requirement matrix $\boldsymbol{A}_{ij}^{\pm}$, transformation matrix $\boldsymbol{B}_{ij}$, and concurrency matrix $\boldsymbol{C}_{ij}^{\pm}$. All the orbital nodes (yellow nodes in Figure 7) are transshipment nodes and therefore we have $\boldsymbol{b}_i = \boldsymbol{0}$ for them.

D. Requirement Matrix $\boldsymbol{A}_{ij}^{\pm}$

The requirement matrix $\boldsymbol{A}_{ij}^{\pm}$ appears in the mass balance constraint in Eq. (7a). In this case, the requirement matrix $\boldsymbol{A}_{ij}^{\pm}$ is only used in one way: ISRU plant requiring maintenance mass. This involves a loop edge modeling facility and there is no requirement matrix for a regular edge. An ISRU system generally includes a power system, a tanker, and a terrain management vehicle as well as a plant. Once an ISRU system is deployed, the maintenance task arises. In this study we assume that the ISRU system is automated or can be teleoperated and that maintenance mass required is a linear function of the total ISRU system mass including ISRU plant and support elements. Hence, if a large ISRU system is deployed, it can produce a large amount of resources but at the same time it requires a large maintenance mass. If α is the proportional constant, then this can be written as:

$$\boldsymbol{A}_{ii}^{+}\boldsymbol{x}_{ii}^{+} = \begin{bmatrix} 1 & 0 \\ \alpha & 1 \end{bmatrix}_{ii}^{+} \begin{bmatrix} \text{plant} \\ \text{maintenance} \end{bmatrix}_{ii}^{+} \tag{14}$$

Downloaded by UNIVERSITY OF TENNESSEE on October 15, 2015 | http://arc.aiaa.org | DOI: 10.2514/6.2013-5414

Note that only relevant variables are shown and the $\boldsymbol{A}_{ii}^{+}$ portion corresponding to the rest variables is an identity matrix. For the α value, we use 10% of the total ISRU system mass per year, which is a conservative assumption.

E. Transformation Matrix $\boldsymbol{B}_{ij}$

The transformation matrix $\boldsymbol{B}_{ij}$ literally models the flow transformation between commodities on an edge as in Eq. (7b). In contrast to the $\boldsymbol{A}_{ij}^{\pm}$ matrix, the transformation matrix $\boldsymbol{B}_{ij}$ is used in many ways.

1. Propulsive Burn

The *propellant mass fraction* is the ratio between mass of the propellant used and the initial mass of the vehicle. From the rocket equation, the propellant mass fraction ϕ_{ij} on edge (i,j) is derived as:

$$\phi_{ij} = 1 - \exp\left(-\frac{\Delta V_{ij}}{I_{\mathrm{sp}} g_0}\right) \tag{15}$$

where ΔV_{ij} is the change in the vehicle's velocity on edge (i,j), I_{sp} is the specific impulse, and g_0 is the standard gravity. Since in this case study we consider chemical propulsion using $\mathrm{LOX/LH_2}$ and NTR, it is suffice to include hydrogen and oxygen as a propellant. Let μ be the mixture ratio of oxidizer to fuel. Then we have

$$\boldsymbol{x}_{ij}^{\pm} = \begin{bmatrix} \text{dry mass} \\ \text{hydrogen} \\ \text{oxygen} \end{bmatrix}_{ij}^{\pm} \qquad \boldsymbol{B}_{ij} = \begin{bmatrix} 1 & 0 & 0 \\ -\dfrac{\phi}{1+\mu} & 1-\dfrac{\phi}{1+\mu} & -\dfrac{\phi}{1+\mu} \\ -\dfrac{\mu\phi}{1+\mu} & -\dfrac{\mu\phi}{1+\mu} & 1-\dfrac{\mu\phi}{1+\mu} \end{bmatrix}_{ij} \tag{16}$$

where "dry mass" includes the commodities other than hydrogen and oxygen.

We assume $I_{\mathrm{sp}} = 450$ [s] for $\mathrm{LOX/LH_2}$ with the mixture ratio of $\mu = 6.0$ and $I_{\mathrm{sp}} = 900$ [s] for NTR. NTR only uses liquid hydrogen but Eq. (16) still holds by setting $\mu = 0$. The ΔV_{ij} value for each edge is listed in Figure 13(a) in Appendix.[29–31] Note that these ΔV values assume high thrust propulsion and the Oberth maneuver if possible. Cells shaded in light red represent aerobraking options, in which ΔV for arrival capture can be saved but the aeroshell is needed that adds to mass. For example, when we transfer from LEO to LMO, it requires 6.12 km/s of ΔV without aerocapture and 3.82 km/s of ΔV with aerocapture.

2. Resource Boil-Off

The cryogenic propellant is lost to boil-off during transport or loiter time. Let r_{bo} be the boil-off rate per unit time. If Δt_{ij} is the duration of edge (i,j), then for any relevant commodity,

$$(1 - r_{\mathrm{bo}})^{\Delta t_{ij}}\, x_{ij}^{+} = x_{ij}^{-} \tag{17}$$

While the thermal environment varies in different locations, we use $r_{\mathrm{bo}} = 0.127\%$ per day ($\simeq 3.81\%$ per month) for liquid hydrogen and $r_{\mathrm{bo}} = 0.016\%$ per day ($\simeq 0.49\%$ per month) for liquid oxygen. The times of flight (TOFs) are provided in Figure 13(b) in Appendix. The TOF for Mars transfer as well as the ΔV are time-variant but we assume a "fast" trajectory of 180 days as with Mars DRA 5.0.

3. Crew Consumables Consumption and Waste Generation

During spaceflight, consumables such as oxygen, water, and food are consumed and waste is generated by the crew. Let c and w be the consumables consumption and the waste generation per unit crew member per unit time, respectively.

If Δt_{ij} is the duration of edge (i,j), then Eq. (7b) holds, where

$$\boldsymbol{x}_{ij}^{\pm} = \begin{bmatrix} \text{crew} \\ \text{consumables} \\ \text{waste} \end{bmatrix}_{ij}^{\pm} \qquad \boldsymbol{B}_{ij} = \begin{bmatrix} 1 & 0 & 0 \\ -c\Delta t_{ij} & 1 & 0 \\ w\Delta t_{ij} & 0 & 1 \end{bmatrix}_{ij} \tag{18}$$

Downloaded by UNIVERSITY OF TENNESSEE on October 15, 2015 | http://arc.aiaa.org | DOI: 10.2514/6.2013-5414

While the net consumption of consumables depends on the ECLSS system and closure rate, we assume that each crew member consumes 0.88 kg of oxygen, 2.90 kg of water, and 1.83 kg of food, and generates 5.61 kg of waste a day. Since the total mass of consumables consumed is equal to the total mass of waste generated ($c = w$), mass is not lost in this transformation. Oxygen is also lost due to leakage from the vehicle. We assume that the spacecraft loses 0.000123 kg of oxygen per vehicle volume in cubic meters per day. These values were derived from the Jet Propulsion Laboratory (JPL) space logistics consumables model, in which the ECLSS closure rate is set to 90%.

4. ISRU Resource Production

ISRU resource production is modeled as a loop edge. Since we assume a linear scalability between the resource production and the ISRU system mass with a proportional constant β, then we have

$$\boldsymbol{x}_{ii}^{\pm} = \begin{bmatrix} \text{plant} \\ \text{resources} \end{bmatrix}_{ii}^{\pm} \qquad \boldsymbol{B}_{ij} = \begin{bmatrix} 1 & 0 \\ \beta & 1 \end{bmatrix}_{ii} \tag{19}$$

Putting this and Eq. (14) together, if an ISRU plant of a mass of x is deployed, it requires maintenance mass of αx and produces resources of βx. For the β value, we assume an ISRU annual production rate of 10 kg per unit ISRU plant mass. For example, if we deploy an ISRU plant of 1,000 kg, it produces resources of 10,000 kg per year. However, this assumption might be somewhat optimistic even if expecting technological advancement in the future. Therefore, a lower ISRU production rate is also attempted later.

5. Mars Surface Stay

Lastly, Mars surface stay is implemented as a loop edge associated with the node of Gale Crater (GC). Since Mars DRA 5.0 assumes 500 days of surface stay, we set the duration of this loop edge to 500 days. Like other edges with duration, resource boil-off and crew consumables consumption/waste generation occur in this loop as well. The only difference with this loop is the transformation of "crew" into "crewRe":

$$\boldsymbol{x}_{ii}^{\pm} = \begin{bmatrix} \text{crew} \\ \text{crewRe} \end{bmatrix}_{ii}^{\pm} \qquad \boldsymbol{B}_{ij} = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}_{ii} \tag{20}$$

Note that it does not represent any actual event or physical process but it just means that the crew is ready for return trip. It is rather a mathematical trick for modeling purposes, serving as a "switch". As stated in Section C, there is a demand of "crewRe" at PAC. In order to satisfy this demand, "crew" must first go to Mars, stay there for 500 days to switch to "crewRe", and come back to Earth. This is the reason why we define "crew" and "crewRe" separately. Without this distinction and a "switch" loop, it would end up satisfying this demand with merely "a round trip to LEO". By implementing this switch at a specific node, we can model a round trip mission to that node.

F. Concurrency Matrix $\boldsymbol{C}_{ij}^{\pm}$

The concurrency matrix $\boldsymbol{C}_{ij}^{\pm}$ represents the constraint between commodities within an edge as in Eq. (7c). The concurrency matrix $\boldsymbol{C}_{ij}^{+}$ is used in three ways: crew transfer vehicle, structural mass for propellant, and aeroshell.

1. Crew Transfer Vehicle

When the crew transfers from node to node, a vehicle to shelter them must travel along with them. Especially for a long-duration interplanetary transfer, they need a "transit habitat" with the appropriate ECLSS system. If we define a minimum vehicle mass per crew member (denoted γ), then we have

$$[\gamma m_{\text{crew}}]_{ij}^{\pm} \leq [m_{\text{vehicle}}]_{ij}^{\pm} \tag{21}$$

Downloaded by UNIVERSITY OF TENNESSEE on October 15, 2015 | http://arc.aiaa.org | DOI: 10.2514/6.2013-5414

This can be rewritten in the form of $\boldsymbol{C}_{ij}^{\pm}$ as:

$$\boldsymbol{x}_{ij}^{\pm} = \begin{bmatrix} \text{crew} \\ \text{vehicle} \end{bmatrix}_{ij}^{\pm} \qquad \boldsymbol{C}_{ij}^{\pm} = \begin{bmatrix} \gamma & -1 \end{bmatrix}_{ii} \tag{22}$$

The total mass of MTH amounts to 37,540 kg and we use this value in a lump as a minimum vehicle mass when the crew transfers on an in-space edge. For other transfer edges such as launch, entry, descent, and ascent, we assume that the crew at least needs the crew exploration vehicle (CEV) that weighs 10,000 kg. The MTH and CEV are designed to accommodate crew of six.

2. Structural Mass for Propellant

Likewise, propellant cannot travel by itself; it needs "structure". The *inert mass fraction* is defined as the ratio of structure mass to structure plus propellant mass:

$$f_{\text{inert}} \equiv \frac{m_{\text{st}}^{+}}{m_{\text{st}}^{+} + m_{\text{pr}}^{+}} \tag{23}$$

so that the ratio of structure mass to propellant mass is:

$$\eta \equiv \frac{m_{\text{st}}^{+}}{m_{\text{pr}}^{+}} = \frac{f_{\text{inert}}}{1 - f_{\text{inert}}} \tag{24}$$

If we have 1 unit of propellant, we must carry at least η units of structure along with it. Then we have

$$\boldsymbol{x}_{ij}^{+} = \begin{bmatrix} \text{propellant} \\ \text{inert} \end{bmatrix}_{ij}^{+} \qquad \boldsymbol{C}_{ij}^{+} = \begin{bmatrix} \eta & -1 \end{bmatrix}_{ij}^{+} \tag{25}$$

which is equivalent to

$$[\eta m_{\text{pr}}]_{ij}^{+} \leq [m_{\text{st}}]_{ij}^{+} \tag{26}$$

We handle inert mass for LOX/LH_2 and NTR separately because these two different propulsion systems cannot share the same structure. In this case study, we assume $f_{\text{inert}} = 0.08$ for LOX/LH_2 (0.15 for ascent/descent) and $f_{\text{inert}} = 0.30$ for NTR.

3. Aeroshell

For Mars arrival or Earth arrival, aerobraking is available with an aeroshell. Since aerobraking is an option, this is modeled as a parallel edge separately from an transfer edge without aerobraking. In total, there are 4 parallel edges between the same end nodes: LOX/LH_2 with/without aerobraking and NTR with/without aerobraking. Let θ denote the aeroshell mass fraction, meaning that when a spacecraft with a mass of 1 performs aerobraking, it must have an aeroshell with a mass of θ. Then we have

$$\boldsymbol{x}_{ij}^{-} = \begin{bmatrix} \text{spacecraft} \\ \text{aeroshell} \end{bmatrix}_{ij}^{-} \qquad \boldsymbol{C}_{ij}^{-} = \begin{bmatrix} \theta & -1 \end{bmatrix}_{ij}^{-} \tag{27}$$

which is equivalent to

$$[\theta m_{\text{sc}}]_{ij}^{-} \leq [m_{\text{as}}]_{ij}^{-} \tag{28}$$

We assume that when performing aerobraking, spacecraft needs aeroshell with a mass of about 15% of the total mass being braked. Note that aerobraking is performed at arrival so that this is an "inflow" concurrency constraint (with a superscript of "$-$").

Downloaded by UNIVERSITY OF TENNESSEE on October 15, 2015 | http://arc.aiaa.org | DOI: 10.2514/6.2013-5414

V. Case Study: Optimization Results and Discussions

This section presents the optimizations results of the case study, followed by the discussions. First we solve the baseline problem and compare the result with the Mars DRA 5.0 scenario, focusing on the difference in the initial mass in low-Earth orbit (IMLEO) and the utilization of infrastructure network. Then we change the problem settings, assumptions, and conditions and perform a sensitivity analysis to find the key drivers and thresholds.

As described earlier in Section III, in this case study the GMCNF LP problem is solved using the CPLEX `cplexlp` function within MATLAB. LP problems do not need much computational effort in general. Using a normal store-bought laptop computer with average specs as of 2013, one instance can be solved within 10 seconds. Therefore it is easy to try many different instances, which greatly helps us in debugging, tuning the problem settings, and performing a sensitivity analysis.

A. Baseline Problem

Downloaded by UNIVERSITY OF TENNESSEE on October 15, 2015 | http://arc.aiaa.org | DOI: 10.2514/6.2013-5414

In the baseline problem we use the same problem setting as the Mars DRA 5.0 scenario. Again the mission objective of Mars DRA 5.0 in a logistics language is to send crew of six and the surface systems to the Martian surface and to bring the crew back to Earth after 500 days of surface stay. We translate this objective into the demand for "habitat" of 51,700 kg at GC and "crewRe" of 600 kg at PAC (assuming 100 kg per crew member). The supply of each commodity is set to infinity at KSC except for "crewRe". In order to make a fair comparison with the Mars DRA 5.0 scenario, we impose an additional condition that we can only use LOX/LH_2 for launch to LEO. In other words, NTR is only allowed in space, which somewhat reflects the political concern about nuclear propulsion.

One important thing to note about this model is that it simulates a static network flow. It does not consider the time evolution of the network flow or the network topology. Therefore, for example, if we obtain the optimization result that utilizes ISRU systems, it assumes that the ISRU system is already deployed and ready to produce resources from the beginning. In other words, it is equivalent to computing a network flow in the "post-deployment" phase. While this model does not consider time evolution, we can arbitrarily define the scope of problem, that is, the time period to be considered. By defining a time period for a network flow, the network flow actually represents the flow rate. If the time period is one day, the network flow is a daily flow; if the time period is one year, the network flow is an annual flow. This can be different from the mission duration, which is about 860 days (180 days on the outbound leg + 500 days on Mars + 180 days on the return leg). While we can choose any length of time for the scope of the problem, it is reasonable to use the lifetime of the ISRU system because we can easily evaluate the life-cycle cost performance of the ISRU system.

Then the question is how long the lifetime of the ISRU system is. It is unlikely that the ISRU system would last for 10 years because, for example, with each passing year, the local ice field around the plant is getting depleted and the rovers transporting regolith to the processing unit must travel greater distances. It might be difficult to keep the constant resource production rate for a long time. Also these working rovers are constantly excavating and transporting regolith and are likely to be worn out quickly. Thus we assume 5 years as the lifetime of the ISRU system, or a reasonable length of time during which the ISRU plant keeps the same pace of resource production.

The network graph (Figure 7) consists of 16 nodes and 357 directed edges including 9 loops in total. There are 14 types of commodities as in Eqs. (12). Using the notations in Section III, we have the number of nodes ($n = 16$), the number of edges ($m = 357$), the number of types of commodities ($k = 14$), and the maximum number of concurrency constraints ($l = 2$). Hence, this problem has 9996 variables, 1652 inequality constraints, and 4998 equality constraints, in addition to 9996 nonnegativity constraints. Again the objective function is the total mass to be launched from Earth (KSC). Since it includes the ISRU plant and maintenance mass (not only the portion for a crew mission to Mars), it serves as a comprehensive performance indicator. Table 1 summarizes the assumptions that have been presented in the previous sections. All the parameter values used in the analysis are presented in Figures 13(a) and 13(b) as well as Table 1.

After about 10 seconds of computation, the problem is successfully solved and the optimal solution turns out to be 2,377,746 kg. The corresponding total IMLEO is 276.0 t. In the Mars DRA 5.0, two reference scenarios are presented: NTR vs. chemical/aerocapture option for in-space transportation.[27] The estimated

Downloaded by UNIVERSITY OF TENNESSEE on October 15, 2015 | http://arc.aiaa.org | DOI: 10.2514/6.2013-5414

Table 1. Summary of assumptions used in the analysis

Parameter	Assumed Value
Scope of problem	5 years
Mission data	
Number of crew	6
Mass per crew member	100 [kg]
Mars Transit Habitat (MTH)	27,540 [kg]
Crew Exploration Vehicle (CEV)	10,000 [kg]
Surface Habitat (SHAB)	51,700 [kg]
Propulsion system	
Chemical rocket (LOX/LH_2)	
Specific impulse I_{sp}	450 [s]
Mixture ratio μ	6.0
Inert mass fraction f_{inert}	0.08 (0.15 for descent/ascent)
Nuclear thermal rocket (NTR)	
Specific impulse I_{sp}	900 [s]
Inert mass fraction f_{inert}	0.30
Boil-off rate r_{bo}	
Hydrogen	0.127% per day ($\simeq$ 3.81% per month)
Oxygen	0.016% per day ($\simeq$ 0.49% per month)
Aeroshell mass fraction θ	15% of total mass being braked
Crew consumables consumption	
Oxygen	0.88 [kg] per crew member per day
Water	2.90 [kg] per crew member per day
Food	1.83 [kg] per crew member per day
Waste generation	5.61 [kg] per crew member per day
Oxygen leakage from vehicle	0.000123 [kg] per vehicle volume per day
ISRU system	
Resource production rate	10 [kg] per plant mass per year
Maintenance mass required	10% of plant mass per year

total IMLEOs are 848.7 t for reference NTR and 1,251.8 t for reference chemical/aerocapture, respectively. The former is equivalent to 9 Ares V launches while the latter is equivalent to 12 Ares V launches. If we simply compare the total IMLEOs, the optimal solution saves 67.5% from the reference NTR scenario and 78.0% from the reference chemical/aerocapture scenario.

Figure 8 shows the major flows of any of the 14 types of commodities of the baseline solution in bold lines on a single network graph. Figure 9 lists the amount of each commodity on the edges used in the baseline solution (bold lines in Figure 8). Putting aside the actual feasibility, it is interesting that the resources produced on Deimos are delivered back to LEO, GTO, and EML2 to be used in the cislunar system. This is probably because Deimos is a little closer to part of the cislunar system than the lunar surface in terms of ΔV (see Figure 13(a)).

Figure 10 shows the flows of "crew" and "crewRe". The flow of crew represents the route that the crew takes in the network. Since we have implemented the concurrency constraint that the vehicle must travel along with the crew, then the route that the crew takes is identical to the route that the vehicle takes. Therefore these figures can simply be interpreted as how the crew goes to Mars and comes back to Earth. Since the crew vehicle headed for Mars is called the Mars transfer vehicle (MTV) in Mars DRA 5.0, it is referred to as the MTV hereafter. As also noted in the figure, the red line represents transport using LOX/LH_2, the blue line represents transport using NTR, and the dashed line represents transport

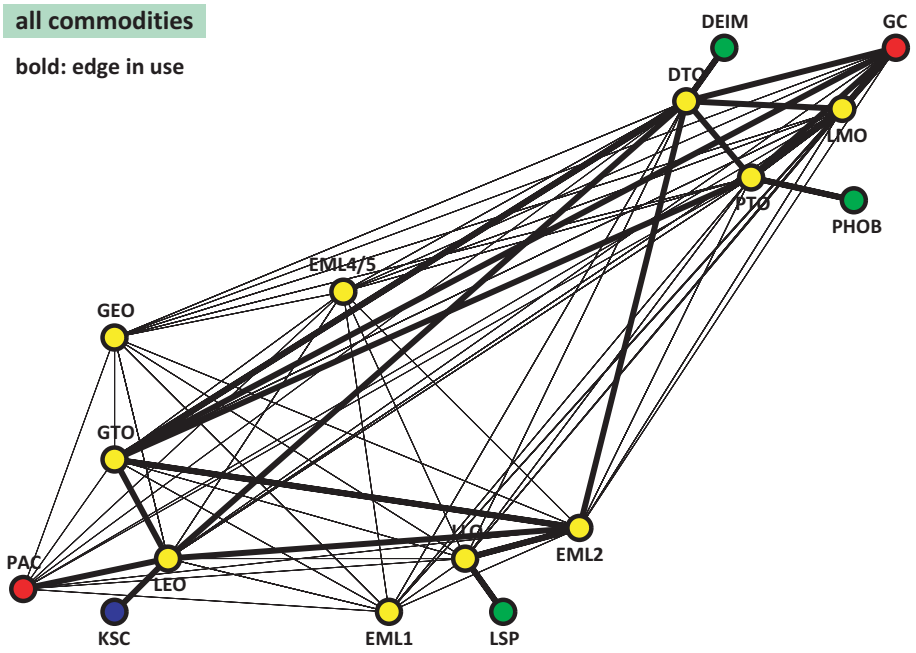


Figure 8. Flows of all commodities of the baseline solution

using aerobraking at arrival. In Figure 10, for example, the solid red line from LEO to GTO represents crew transport from LEO to GTO using LOX/LH_2 (all-propulsive) while the dashed blue line from DTO to GC represents crew transport from DTO to GC using NTR with aerobraking at arrival.

For the outbound leg, after launch to low-Earth orbit (LEO), the MTV first stops by at geostationary transfer orbit (GTO) and then is injected into the trans-Mars trajectory. At Mars arrival, the MTV also stops by at Deimos transfer orbit (DTO), followed by atmospheric entry. The crew spends 500 days on the surface. On the other hand, the crew vehicle returning to Earth is called the Earth return vehicle (ERV). For the return leg, after ascent to low Mars orbit (LMO), the ERV stops by at Phobos transfer orbit (PTO) and DTO like a "local train". Then the ERV is injected to trans-Earth trajectory to LEO, followed by atmospheric reentry and Pacific Ocean splashdown. As for the propulsion system, NTR is used only for the entry, descent, and landing (EDL) to GC and the rest uses LOX/LH_2. Also, aerobraking is used for orbit capture in DTO and entry to Martian atmosphere for the outbound leg, and for orbit capture in LEO and reentry to Earth atmosphere for the return leg. Therefore it turns out that except for EDL to GC, LOX/LH_2 is used throughout the mission.

Table 2 lists the result of ISRU at each surface node. We can see the ISRU plant is carried to all the surface nodes, which means that the ISRU plant is deployed in all the ISRU candidate locations. Note that we assume that water ice is available on the Moon, Deimos, and Phobos while oxygen is produced via Sabatier reaction on Mars. Putting aside the actual resource availabilities in these locations, it turns out that the largest ISRU plant is deployed on Deimos, weighing 4,856 kg, the second largest is on the Moon, weighing 4,821 kg, the third largest is on Phobos, weighing 2,504 kg, and lastly the smallest is deployed on Mars, weighing 1,057 kg. The total amount of resources produced and maintenance mass required for 5 years are also listed in the table. Again, this production rate might be too aggressive. Other instances with lower production rates are described later in the subsequent section. One interesting thing is that since all the ISRU plants are delivered through GTO, it serves as a hub in the network. This, along with the fact that the MTV's long trip to Mars also starts out from GTO, implies the importance of GTO in the network.

To summarize, it is found that the optimal solution makes the most use of ISRU in all the candidate locations. The ISRU plants are delivered to those locations as in Table 2, and in turn the resources produced are delivered to orbital nodes. The MTV stops by at each orbital node to refuel itself, instead of directly flying to Mars and back to Earth. One thing to note is that the solution is just a network flow and it does not

Downloaded by UNIVERSITY OF TENNESSEE on October 15, 2015 | http://arc.aiaa.org | DOI: 10.2514/6.2013-5414

from	to	prop.	aero?	vehicle	habitat	crew	crewRe	hydrogen	oxygen	water	food	waste	inertLOX	inertNTR	aeroshell	plant	mainte.
KSC	LEO	LOX/LH_2	NO	10000	51700	600	0	0	0	0	9509	34	182561	0	1768	13239	6619
LEO	GTO	LOX/LH_2	NO	37540	51700	600	0	0	0	0	9487	34	16648	0	27977	13239	6619
LEO	PAC	LOX/LH_2	YES	10000	0	0	600	0	0	0	0	34	2	0	1595	0	0
GTO	EML2	LOX/LH_2	NO	0	0	0	0	0	0	0	0	0	1589	0	2936	4821	2411
GTO	DTO	LOX/LH_2	YES	37540	0	600	0	0	0	0	1987	6059	15963	0	25445	4856	2428
GTO	PTO	LOX/LH_2	YES	0	45565	0	0	0	0	0	5523	0	7196	0	9306	2504	1252
GTO	GC	LOX/LH_2	YES	0	6135	0	0	0	0	0	0	0	279	0	1200	1057	529
LSP	LLO	LOX/LH_2	NO	0	0	0	0	23550	11142	0	0	0	10312	0	0	0	0
LLO	LSP	LOX/LH_2	NO	0	0	0	0	0	0	0	0	0	10322	0	0	4821	2411
LLO	EML2	LOX/LH_2	NO	0	0	0	0	21665	0	0	0	0	2209	0	0	0	0
EML2	LEO	LOX/LH_2	YES	0	0	0	0	17843	0	0	0	0	1710	0	2933	0	0
EML2	GTO	LOX/LH_2	NO	0	0	0	0	3137	0	0	0	0	345	0	0	0	0
EML2	LLO	LOX/LH_2	NO	0	0	0	0	0	0	0	0	0	2231	0	0	4821	2411
DTO	LEO	LOX/LH_2	YES	37540	0	0	600	0	107070	35	0	6059	14707	0	24902	0	0
DTO	GTO	LOX/LH_2	YES	0	0	0	0	6000	55799	3132	0	0	8059	0	10948	0	0
DTO	EML2	LOX/LH_2	NO	0	0	0	0	0	3473	0	0	0	493	0	0	0	0
DTO	DEIM	LOX/LH_2	NO	0	0	0	0	0	0	0	0	0	42847	0	0	4856	2428
DTO	LMO	LOX/LH_2	YES	27540	0	0	0	0	0	0	0	0	1	0	4131	0	0
DTO	GC	NTR	YES	10000	0	600	0	0	0	0	0	34	0	8	1596	0	0
DEIM	DTO	LOX/LH_2	NO	0	0	0	0	20450	174080	6316	0	0	42804	0	0	0	0
PTO	DTO	LOX/LH_2	NO	37540	0	0	600	0	75551	0	0	34	7405	8	16174	0	0
PTO	PHOB	LOX/LH_2	NO	0	0	0	0	0	0	0	0	0	22094	0	0	2504	1252
PTO	LMO	LOX/LH_2	YES	0	0	0	0	1848	0	17	11	0	162	0	306	0	0
PTO	LMO	NTR	YES	0	0	0	0	0	11092	0	0	0	0	1	1664	0	0
PTO	GC	LOX/LH_2	YES	0	45565	0	0	8049	0	0	5501	0	701	0	8972	0	0
PHOB	PTO	LOX/LH_2	NO	0	0	0	0	11737	97618	35	0	0	22072	0	0	0	0
LMO	PTO	LOX/LH_2	NO	37540	0	0	600	0	0	0	0	34	1125	9	17837	0	0
LMO	GC	LOX/LH_2	YES	0	0	0	0	0	0	0	0	0	7762	0	1164	0	0
GC	LMO	LOX/LH_2	NO	10000	0	0	600	0	0	0	0	34	8733	8	12920	0	0

Figure 9. Commodity delivered through each edge in the baseline solution

Table 2. ISRU at each surface node

	LSP	DEIM	PHOB	GC
ISRU plant deployed	4,821 [kg]	4,856 [kg]	2,504 [kg]	1,057 [kg]
Resources produced (5 years)	241,074 [kg]	242,799 [kg]	125,197 [kg]	52,858 [kg]
Maintenance required (5 years)	2,411 [kg]	2,428 [kg]	1,252 [kg]	529 [kg]

indicate any concrete operation such as depots. The flow just converges and diverges at each node. Nodes at which the flow merges do not necessarily suggest fuel depots. Another interpretation of flow convergence and divergence could be that two vehicles rendezvous, fly together to some point, and separate. This is a "pickup bus" style.

One possible scenario that can be built from the flow information is shown in Figure 10. There are two resource depots in this scenario: one at GTO and the other at DTO. First the MTV is launched from KSC to LEO using NTR and rendezvous with an orbital transfer vehicle (OTV) at LEO. The OTV takes the MTV to a resource depot at GTO and the MTV refuels itself at the depot. This OTV and depot are maintained with the lunar resources produced at LSP as well as the Deimos resources produced at DEIM. Then the MTV is injected into trans-Mars trajectory and after 180 days it uses aerocapture to insert itself into DTO to rendezvous with the other resource depot. This depot is supplied by Deimos and Phobos resources. Having refueled at the depot, the CEV performs EDL to GC using NTR with aerobraking. For the return leg, the crew ascends to LMO and gets aboard the ERV. Then another OTV takes the ERV to PTO. At PTO, the OTV refuels itself by Phobos resources or the ERV switches to yet another OTV. This is still disputable but the ERV somehow utilizes Phobos resources to get to the DTO depot, which is the same one as used in the outbound leg. Having refueled at this depot, the ERV is injected into trans-Earth trajectory and after 180

Downloaded by UNIVERSITY OF TENNESSEE on October 15, 2015 | http://arc.aiaa.org | DOI: 10.2514/6.2013-5414

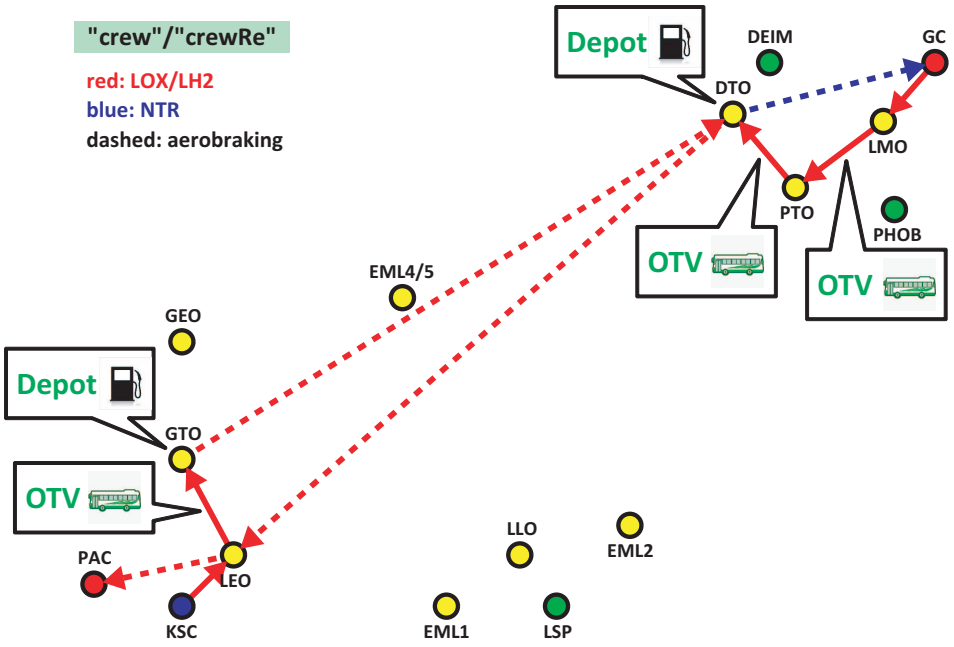


Figure 10. Flows of "crew" and "crewRe" and example scenario for the baseline solution

days it uses aerocapture to insert itself into LEO. Finally the crew aboard the CEV performs a reentry and splashes down in the Pacific Ocean (PAC). This is one example scenario that represents the network flow result of the baseline solution.

In Mars DRA 5.0, the reference NTR scenario is much better than the reference chemical/aerocapture scenario in terms of IMLEO, whereas in the optimal solution, while the chemical (LOX/LH_2) rocket is mainly used, the IMLEO is cut as much as 67.5% from the DRA 5.0 NTR scenario. As we have seen, the biggest reason is obviously ISRU. Since ISRU provides much oxygen and some hydrogen, it can be inferred that LOX/LH_2 is likely to take advantage of ISRU because it heavily uses oxygen.

B. Variant Problems

This section attempts several different settings and parameters. Based on the baseline problem presented in the previous section, we solve variant problems that have different settings or parameters in: (1) propulsion system, (2) ISRU availability, (3) boil-off rate, (4) ISRU system lifetime, and (5) ISRU resource production rate. We especially focus on how the total mass to be launched from KSC and the network topology vary with these factors.

1. Propulsion System

First we solve the problems with some limitations on propulsion systems. The two reference scenarios in Mars DRA 5.0 do not allow the combination use of chemical and nuclear thermal propulsions while the baseline solution in the previous section uses both. Therefore, we look at three cases: (1) LOX/LH_2 only with aerobraking option, (2) NTR only with aerobraking option, and (3) LOX/LH_2 and NTR without aerobraking option. Note that other parameters and assumptions remain the same.

Table 3 compares the results of these variants with the baseline solution. These variants are essentially the problems with limited options on propulsion systems so that the results never improve. As opposed to the Mars DRA 5.0 scenarios in which NTR is preferred to chemical, it turns out that the second best is (1) LOX/LH_2 with aerobraking. This is probably because ISRU can provide much oxygen, which greatly reduces the propellants that must be brought from Earth. NTR is two times better than LOX/LH_2 in specific impulse (I_{sp}) and this is the primary reason why NTR is preferred in Mars DRA 5.0. However, this

Downloaded by UNIVERSITY OF TENNESSEE on October 15, 2015 | http://arc.aiaa.org | DOI: 10.2514/6.2013-5414

result shows that ISRU oxygen production helps reduce the initial mass at KSC more. On the other hand, the third variant without aerobraking option has the largest initial mass, which is more than twice that of the baseline scenario. In other words, the aerobraking option makes the most important contribution in saving the initial mass at KSC.

Table 3 also shows the size of ISRU plant deployed at each node. The first scenario with chemical/aerobrake results in almost the same allocation of the ISRU plants as in the baseline scenario, which makes sense because the baseline scenario relies mostly on LOX/LH_2 and ISRU oxygen. Figure 14(b) shows the path that the crew takes to Mars and back to Earth, which is nearly the same as the one in the baseline (Figure 14(a)). The second scenario with NTR/aerobrake does not utilize the lunar resources while it heavily relies on Phobos and Deimos ISRU. Figure 14(c) illustrates the crew's route. This scenario tends to take a straightforward route except for a "local train" refueling in the Martian system. Lastly, the third scenario without aerobraking option turns out to heavily utilize the ISRU resources, especially from the lunar and Deimos surfaces. The outbound and return routes are illustrated in Figure 14(d), which is somewhat different from the previous ones. For the outbound leg, the MTV transfers from GTO to EML2 before injection into trans-Mars trajectory. In this scenario, a resource depot should be located in EML2 instead of GTO. It appears to be a detour but it actually is not as costly as it looks because the Oberth maneuver is available for trans-Mars injection from EML2.

Downloaded by UNIVERSITY OF TENNESSEE on October 15, 2015 | http://arc.aiaa.org | DOI: 10.2514/6.2013-5414

2. ISRU Availability

In the baseline problem, we assume that ISRU is available on the Moon, Deimos, Phobos, and Mars. However, the resource availability is not guaranteed in all these locations. Especially, ISRU on Phobos and Deimos is controversial although the evidence that there is likely to be water ice in the interior of Deimos and/or Phobos is provided in the literature.[23] Another story leading to poor ISRU availability is that the ISRU technology including excavating and transporting raw materials is not as effective as expected or even infeasible so that we cannot greatly rely on ISRU. This section examines the following three cases: (4) ISRU available on the Moon and Mars, (5) ISRU available on the Moon only, (6) ISRU available on Mars only, and (7) ISRU not available. Again note that other parameters and assumptions are not changed from the baseline problem.

Likewise, Table 3 compares these three scenarios with the baseline. Again these variants limit the ISRU availabilities so that the results never improve. In scenario (4) in which ISRU is available on the Moon and Mars and scenario (5) in which ISRU is only available on the Moon, the initial mass at KSC does not degrade very much despite the fact that the baseline scenario uses a substantial amount of Deimos and Phobos resources. Instead, the lunar ISRU plant is much upgraded so as to compensate the loss of Deimos and Phobos contributions. Figure 14(e) and 14(f) illustrate the same route except that LOX/LH_2 is used for Mars ascent in scenario (4). Similarly to scenario (3) in Figure 14(d), the outbound leg passes through EML2. This implies that EML2 could be an alternative to GTO as a cislunar gateway. NASA has been focusing on the Flexible Path idea and exploration platform at EML2 and interestingly this result is consistent with that idea while it is merely a result that automatically comes out from optimization. In scenario (6) in which ISRU is only available on Mars and scenario (7) in which no ISRU is available, the initial mass at KSC drastically increases. Figures 14(g) and 14(h) illustrate the same route except that LOX/LH_2 is used for Mars ascent in scenario (6). As can be easily understood, as the ISRU availability decreases, the complexity of network flow decreases and the path becomes more "direct" and tends toward NTR. This makes intuitive sense because oxygen-rich ISRU helps make LOX/LH_2 more effective than NTR. To summarize, it is found that ISRU availability is one of the key drivers to improve the initial mass at KSC.

3. Boil-Off Rate

In the baseline problem, we assume that for liquid hydrogen the boil-off rate is 0.127% per day (3.81% per month) while that of liquid oxygen is 0.016% per day (0.49% per month). Boil-off of cryogenic propellants in space is caused by heating from solar and other sources. This could be mitigated by technological solutions as well as system-level planning solutions. One system-level solution is reductions in the standby time of the cryogenic storage by a just-in-time (JIT) delivery to a customer. However, when it comes to Mars exploration, we cannot avoid a long-duration expedition. For example, if we need to carry hydrogen along with the 180-day interplanetary flight, the boil-off would be over 20%. Therefore planning solutions would not replace the need for efficient technological storage solutions. While a passive insulation system to effectively

store cryogenic fluids could mitigate the boil-off, it is possible to achieve zero boil-off (ZBO) using an active thermal control system. Thus we look at the ZBO scenario, in which we assume an ideal 0% boil-off rate for both hydrogen and oxygen.

Table 3 compares the zero boil-off (ZBO) scenario with the baseline. It turns out that the initial mass at KSC improves 2% from the baseline. This improvement is not very big but the network topology changes; the lunar ISRU is not utilized at all and instead the plants on Deimos and Phobos are upgraded. As far as this result is concerned, the zero boil-off technology does not make a big difference in the initial mass at KSC. However, in this model we do not consider the standby time of the cryogenic storage, that is, we assume a just-in-time delivery. This assumption might make the objective function somewhat less sensitive to boil-off rate.

Table 3. Summary of the solutions with various settings

Scenario	Initial Mass	ISRU Plant			
	at KSC ($\mathcal{J}$)	LSP	DEIM	PHOB	GC
Baseline Scenario	2,377,746 [kg]	4,821 [kg]	4,856 [kg]	2,504 [kg]	1,057 [kg]
Propulsion System					
(1) Chemical/Aerobrake	2,378,368 [kg]	4,822 [kg]	4,857 [kg]	2,510 [kg]	1,057 [kg]
(2) NTR/Aerobrake	3,797,386 [kg]	0 [kg]	27,088 [kg]	12,711 [kg]	208 [kg]
(3) Chemical/NTR	5,232,681 [kg]	27,794 [kg]	52,315 [kg]	8,272 [kg]	806 [kg]
ISRU Availability					
(4) ISRU on Moon/Mars	2,721,529 [kg]	21,138 [kg]	0 [kg]	0 [kg]	1,057 [kg]
(5) ISRU on Moon only	2,912,353 [kg]	27,161 [kg]	0 [kg]	0 [kg]	0 [kg]
(6) ISRU on Mars only	5,562,650 [kg]	0 [kg]	0 [kg]	0 [kg]	910 [kg]
(7) No ISRU	6,564,098 [kg]	0 [kg]	0 [kg]	0 [kg]	0 [kg]
Boil-Off Rate					
(8) Zero boil-off	2,331,486 [kg]	0 [kg]	6,796 [kg]	4,174 [kg]	1,057 [kg]

4. ISRU System Lifetime

In the baseline problem we make an arbitrary assumption that the ISRU system lifetime (during which the ISRU plant keeps the constant resource production rate) is 5 years. However, if it is shorter/longer, that might be a different story. For example, if the lifetime is very short, the amount of the total resources produced will be low with respect to the ISRU plant mass, which implies that it is not worth the investment. Therefore we change the ISRU lifetime between 1 and 10 years and look at how the initial mass at KSC and the ISRU plant mass in each location vary with the lifetime.

Figure 11 plots the initial mass at KSC (to the left axis) and the ISRU plant mass in each location (to the right axis) with respect to the assumed ISRU system lifetime. If the ISRU system only lasts for a year, the lunar ISRU is not invested at all while the other three locations should be utilized. If the ISRU system lasts for two years, the lunar ISRU starts to be used and the flow network topology changes. If the ISRU lifetime is longer than two years, as the lifetime gets longer, the initial mass at KSC goes down, converging to around 2,200,000 kg. This is apparent because as the lifetime gets longer, the amount of lifecycle resource production increases with respect to the same plant mass to be delivered. The plant mass in each location also decreases. However, it does not mean a reduction in investments. Since the ISRU plant produces more resources if it lasts longer, a smaller plant can keep the same amount of resource supply and the network topology does not basically change. To summarize, unless the lifetime is very short, ISRU can provide great value.

Downloaded by UNIVERSITY OF TENNESSEE on October 15, 2015 | http://arc.aiaa.org | DOI: 10.2514/6.2013-5414

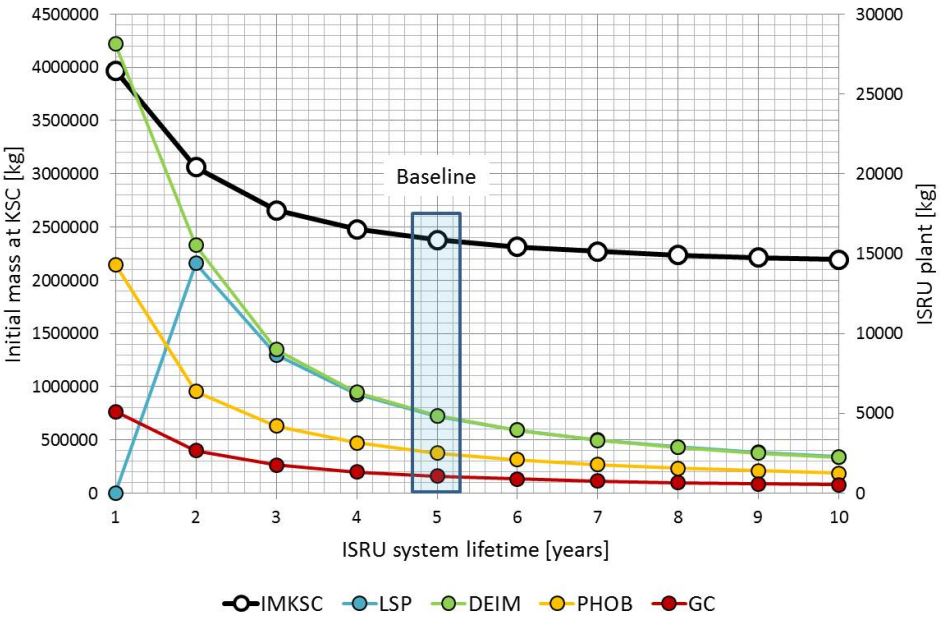


Figure 11. Initial mass at KSC and ISRU plant with respect to ISRU lifetime

5. ISRU Resource Production Rate

Lastly, we vary the ISRU resource production rate. In the baseline problem we assume an ISRU annual production rate of 10 kg per unit plant mass. However, we should know how sensitive the result is to the production rate, particularly in the case of a lower productivity for some reason. It is easy to imagine that if the production rate is below a certain threshold, ISRU is no longer beneficial and we should rather go without ISRU. We change the ISRU annual resource production per plant mass between 0 and 10 kg. Obviously a production rate of 0 converges to scenario (7), in which no ISRU is allowed. Also, since a low production rate is equivalent to a short lifetime in the context of this model, we can expect that the result shows a similar trend to the previous case.

Figure 12 plots the initial mass at KSC (to the left axis) and the ISRU plant mass in each location (to the right axis) with respect to the annual resource production per plant mass. For the annual production rates between 5 and 10, the results are straightforward; the initial mass at KSC simply increases and the ISRU plant in each location becomes upgraded while keeping the same network topology. At the annual production rate of 4, the lunar ISRU discontinuously drops to zero, meaning that if the production rate is below a certain threshold between 4 and 5, the lunar ISRU should not be used at all. Similarly, if the annual production rate falls below another threshold between 2 and 3, ISRU on Deimos becomes worthless while the resulting network highly relies on Phobos ISRU. For the annual production rate below 2, both Phobos and Mars ISRU become downgraded and finally at some point between 0.25 and 0.5, the cost exceeds the benefit. As for the propulsion system, LOX/LH_2 is still mainly used at a production rate of 4 where the lunar ISRU drops to zero while the main propulsion system is switched to NTR at a production rate of 2 where the Deimos ISRU drops to zero and the crew path becomes more direct, which is very intuitive because as the ISRU resources we can utilize decrease, it makes sense to go straight to the destination and come straight back home using a rocket with a higher I_{sp}.

To summarize, as the ISRU resource production rate decreases from 10 kg/kg/year of the baseline, the initial mass at KSC monotonically increases as a matter of course whereas the ISRU plant mass in each location discontinuously drops to zero at a certain threshold, meaning that below the threshold the cost exceeds the benefit. With the other settings and assumptions of the baseline problem, the lunar ISRU is worth the investment for the production rate of 5 or greater, the Deimos ISRU for 3 or greater, and the Phobos and Mars ISRU for 0.5 or greater.

Downloaded by UNIVERSITY OF TENNESSEE on October 15, 2015 | http://arc.aiaa.org | DOI: 10.2514/6.2013-5414

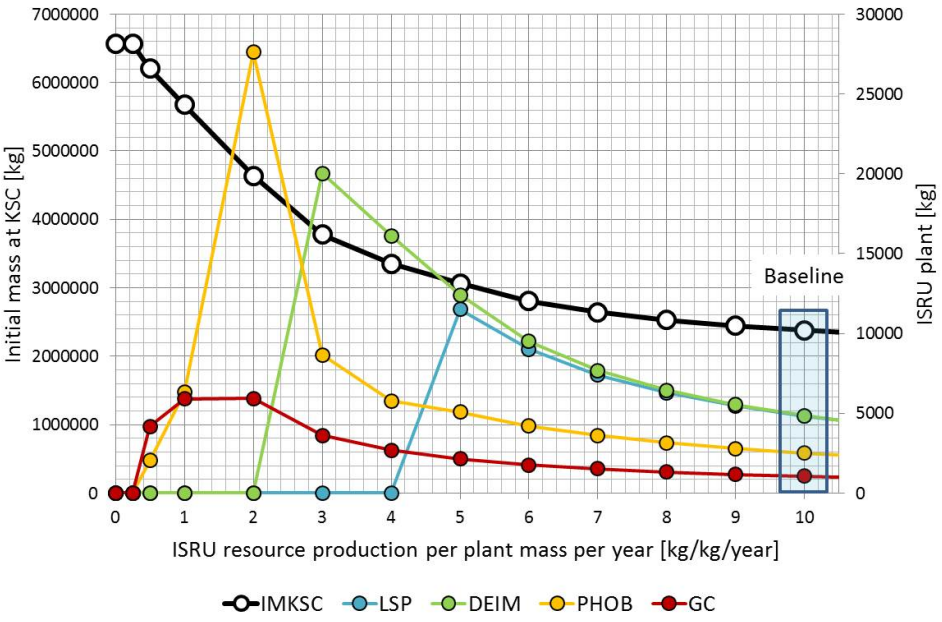


Figure 12. Initial mass at KSC and ISRU plant with respect to ISRU production rate

C. Discussions

1. *Summary of Key Results*

First we solved the baseline problem with settings and assumptions presented in Figures 13(a) and 13(b) as well as Table 1. The baseline solution turned out to save 67.5% from the Mars DRA 5.0 reference NTR scenario in terms of the initial mass in LEO (IMLEO). In the baseline solution, all the four ISRU candidates are utilized and as opposed to the reference NTR scenario, the LOX/LH_2 propulsion is mainly used throughout the mission, taking maximum advantage of oxygen-rich ISRU. One possible scenario that can be built from the result is given in Figure 10, in which two resource depots are located at GTO and DTO as gateways of the cislunar and Martian systems, respectively, and orbital transfer vehicles (OTVs) run in each system like a pickup bus.

Based on this baseline solution, we conducted several what-if analyses with different settings to see the effect of each factor. Table 3 summarizes the results of theses variants as well as the baseline solution. The first three scenarios investigated different propulsion system options. Since oxygen was assumed to be available from ISRU, LOX/LH_2 turned out to be preferred to NTR. Also it was found that the aerobraking option makes a great contribution in reducing the initial mass at KSC. The second four variants attempted different ISRU availabilities. As long as the lunar ISRU is available, the initial mass at KSC does not degrade very much while in the cases of ISRU only available on Mars and of no ISRU, the initial mass is drastically increased, which obviously means that ISRU on the Moon greatly helps to reduce the launch requirement. As the ISRU availabilities decrease, the network complexity decreases concurrently. For the zero boil-off scenario, we did not see a big improvement in the initial mass within the scope of this case study. Also we can see from Figures 11 and 12 that as the ISRU resource production increases/decreases, the network topology changes at several thresholds.

2. *Limitations*

The limitations of the GMCNF LP model primarily arise in three areas: risk analysis, model linearity, and time evolution of network flow/topology. The current model assumes that all transports occur with certainty and all demands are purely deterministic. The risks of node and edge failures are not treated in the model. Robustness optimization needs to be implemented to take into account those risks.

Downloaded by UNIVERSITY OF TENNESSEE on October 15, 2015 | http://arc.aiaa.org | DOI: 10.2514/6.2013-5414

The second limitation of the model is model linearity. In the GMCNF model developed in this study, both the objective function and the constraints are all written in the linear form. While nonlinearity in the predetermined parameters can be built in such as ΔV and TOF, nonlinear objective function or constraints on the flow cannot be handled in this model. For example, we assume a linear scalability between the size of an ISRU plant and the amount of resources produced. If we perform subsystem-level detailed design and planning, we would need a high fidelity model. At the same time, however, there is a great advantage in linear formulation, that is, LP problems can be solved quickly so that we can solve many instances, which helps make trade studies more efficient.

The third limitation of the model is time evolution of network flow/topology. This model simulates a static network flow for a given snapshot of supply/demand and therefore it does not consider the time evolution of the network flow or the network topology. It determines the optimal flow such that it only satisfies the mass balance at each node. Hence, it cannot capture literally the "flow" of events. For example, when the ISRU plant is delivered to the lunar surface at the beginning, obviously we cannot utilize the lunar resources. However, this model does not know that the deployment of the plant must happen before the lunar resources become available so that the contradiction is inherent in the model that the delivery of the lunar plant utilizes the resources produced by itself. Therefore, one possible future work is to implement the "assembly" phase optimization in the model. Considering these limitations, this model is probably more useful for high level system trades such as mission architectures and infrastructure concepts.

Downloaded by UNIVERSITY OF TENNESSEE on October 15, 2015 | http://arc.aiaa.org | DOI: 10.2514/6.2013-5414

VI. Conclusions and Future Work

A. Summary

With the advent of a new era of human space exploration, the research presented in this paper originates from the question of what the next-generation space logistics paradigm should be. The past studies on space logistics have been mainly focused on a "vehicle" perspective with the arbitrarily determined logistics network. The objective of this paper was to develop a comprehensive graph-theoretic modeling framework to quantitatively evaluate and optimize space exploration logistics from a "network" perspective.

In an attempt to create such a modeling framework, we developed a novel network flow model referred to as the generalized multi-commodity network flow (GMCNF) model. On top of the classical network flow theory and its limitations, the GMCNF model proposed in this paper introduces three types of matrix multiplications (requirement, transformation, and concurrency), and also allows loop edges associated with nodes (graph loops) and multiple edges between the same end nodes (multigraph). With this modification, the model can handle multiple commodities that interact with each other in the form of requirement at nodes, transformation on edges, and concurrency within edges. A linear programming (LP) formulation of the GMCNF model was described in preparation for the case study.

As a case study to demonstrate the GMCNF LP model, we applied it to human exploration of Mars and compared the results with NASA's Mars DRA 5.0. First we solved the baseline problem with a demand that is equivalent to that of the Mars DRA 5.0 scenario. It was found that the solution saves 67.5% from the Mars DRA 5.0 reference scenario in terms of the initial mass in low-Earth orbit (IMLEO) primarily because chemical (LOX/LH_2) propulsion is used along with oxygen-rich ISRU. We also presented one possible scenario with two "gateway" resource depots at GTO and DTO with orbital transfer vehicles (OTVs) running in the cislunar and Martian systems. Then we solved variant problems with different settings to see the effect of each factor. Findings include: taking advantage of oxygen-rich ISRU, LOX/LH_2 is preferred to nuclear thermal rocket (NTR), the aerobraking option as well as ISRU availability on the Moon make great contributions in reducing the initial mass at KSC, the zero boil-off (ZBO) does not make a big difference in the initial mass within the scope of this case study, and as the ISRU resource production decreases, ISRU in each location becomes worthless at a certain threshold and the network topology changes toward direct paths using NTR.

The GMCNF model could be applied to any network flow problem (or any problem that is translatable into a network flow problem) in which multiple commodities interact with each other in the form of requirement at nodes, transformation on edges, and concurrency within edges. Also, allowing multiple edges between the same end nodes enables built-in trades between multiple options while allowing graph-loops associated with nodes enables modeling of a facility that requires other commodities and transforms a commodity into another. As discussed in Section V, the limitations of the GMCNF model primarily arise in

three areas: risk analysis, model linearity, and time evolution of network flow/topology.

B. Recommendations for Future Work

While one obvious future work is to implement the GMCNF model in SpaceNet as a front-end tool for network auto-generation, the recommendations for mode general future work are based on the current limitations. One possible way to add robustness optimization to the model is to introduce probabilistic failure of node/edge in any way. In that case, however, the model would not keep its current linear form any longer and therefore this will relate to the second limitation as well.

Removing the limitation of model linearity seems straightforward because a number of solution methods and solvers are available for nonlinear programming (NLP) problems and mixed integer nonlinear programming (MINLP) problems. However, it should be noted that the effectiveness of LP would be sacrificed in return for the generality in the problem structures.

Removing the third limitation seems more challenging. The GMCNF model presented in this thesis can only handle the optimization between two points. As seen in the example of ISRU plant deployment, the current model counts chickens before they hatch. If the temporal dimension is somehow embedded, this contradiction inherent in the current model will be solved. One possible solution is the time-expanded network, in which a set of nodes of the underlying static network is copied for each discrete time step (building a time layer). However, it will raise another issue that due to the size of the time-expanded network that is generated, the problem can become quite large with millions of variables and constraints, which requires much more computational effort. It might be worth exploring a smarter solution to this "back to the future" paradox.

Downloaded by UNIVERSITY OF TENNESSEE on October 15, 2015 | http://arc.aiaa.org | DOI: 10.2514/6.2013-5414

Acknowledgments

The research described in this paper was carried out at the Massachusetts Institute of Technology (MIT) and was supported in part by NASA Jet Propulsion Laboratory (JPL) and in part by Center for Complex Engineering Systems (CCES) at King Abdulaziz City for Science and Technology (KACST) and MIT. The author wishes to acknowledge the support of Thomas Coffee and Paul Grogan at MIT.

Appendix

Figure 13 shows the ΔV values [km/s] and TOFs [days] used in the analysis. Figure 14 shows the flows of "crew" and "crewRe" for the baseline scenario and the variants (1) through (7).

Downloaded by UNIVERSITY OF TENNESSEE on October 15, 2015 | http://arc.aiaa.org | DOI: 10.2514/6.2013-5414

ΔV [km/s]	KSC	PAC	LEO	GTO	GEO	LSP	LLO	EML1	EML2	EML4/5	DTO	DEIM	PTO	PHOB	LMO	GC
KSC			9.50													
PAC																
LEO		9.50		2.50	4.33		4.04	3.77	3.43	3.97	4.92		5.22		6.12	10.22
		0.00									3.82		3.82		3.82	3.82
GTO		12.00	2.50		1.83		1.54	1.27	0.93	1.47	2.42		2.72		3.62	7.72
		0.23	0.23								1.32		1.32		1.32	1.32
GEO		13.83	4.33	1.83			2.05	1.38	1.47	1.71	3.00		3.30		4.20	8.30
		2.06	2.06								1.90		1.90		1.90	1.90
LSP							1.87									
LLO		13.54	4.04	1.54	2.05	1.87		0.64	0.64	0.98	3.10		3.40		4.30	8.40
		1.31	1.31								2.00		2.00		2.00	2.00
EML1		13.27	3.77	1.27	1.38		0.64		0.14	0.33	1.84		2.14		3.04	7.14
		0.77	0.77								0.74		0.74		0.74	0.74
EML2		12.93	3.43	0.93	1.47		0.64	0.14		0.34	1.84		2.14		3.04	7.14
		0.33	0.33								0.74		0.74		0.74	0.74
EML4/5		13.47	3.97	1.47	1.71		0.98	0.33	0.34		2.13		2.43		3.33	7.43
		0.84	0.84								1.03		1.03		1.03	1.03
DTO		14.42	4.92	2.42	3.00		3.10	1.84	1.84	2.13		0.70	0.30		1.20	5.30
		1.10	1.10	1.10									0.00		0.00	0.00
DEIM											0.70					
PTO		14.72	5.22	2.72	3.30		3.40	2.14	2.14	2.43	0.30			0.50	0.90	5.00
		1.40	1.40	1.40											0.00	0.00
PHOB													0.50			
LMO		15.62	6.12	3.62	4.20		4.30	3.04	3.04	3.33	1.20		0.90			4.10
		2.30	2.30	2.30												0.00
GC															4.10	

Aeroshell needed for aerobraking

(a) ΔV values, km/s

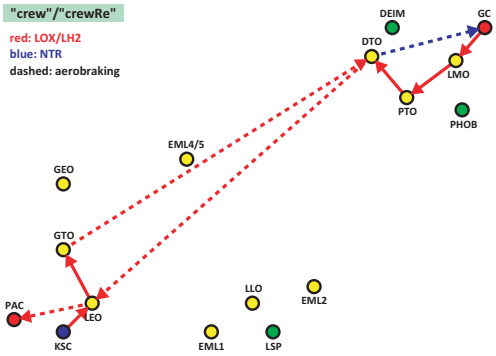
TOF [days]	KSC	PAC	LEO	GTO	GEO	LSP	LLO	EML1	EML2	EML4/5	DTO	DEIM	PTO	PHOB	LMO	GC
KSC			1													
PAC																
LEO		1		1	1		4	3	4	4	180		180		180	180
		1									180		180		180	180
GTO		1	1		1		4	3	4	4	180		180		180	180
		1	1								180		180		180	180
GEO		1	1	1			4	3	4	4	180		180		180	180
		1	1								180		180		180	180
LSP							1									
LLO		4	4	4	4	1		1	1	4	180		180		180	180
		4	4								180		180		180	180
EML1		3	3	3	3		1		1	4	180		180		180	180
		3	3								180		180		180	180
EML2		4	4	4	4		1	1		4	180		180		180	180
		4	4								180		180		180	180
EML4/5		4	4	4	4		4	4	4		180		180		180	180
		4	4								180		180		180	180
DTO		180	180	180	180		180	180	180	180		1	1		1	1
		180	180	180									1		1	1
DEIM											1					
PTO		180	180	180	180		180	180	180	180	1			1	1	1
		180	180	180											1	1
PHOB													1			
LMO		180	180	180	180		180	180	180	180	1		1			1
		180	180	180												1
GC															1	

Aeroshell needed for aerobraking

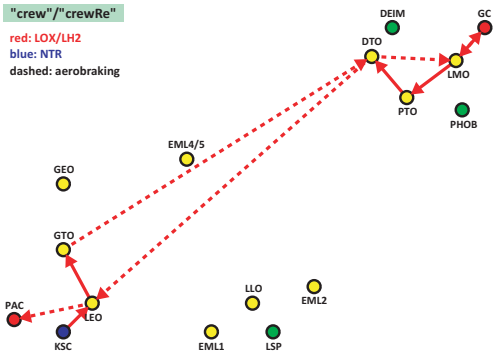
(b) Time of flight (TOF), days

Figure 13. ΔV values and times of flight used in the analysis

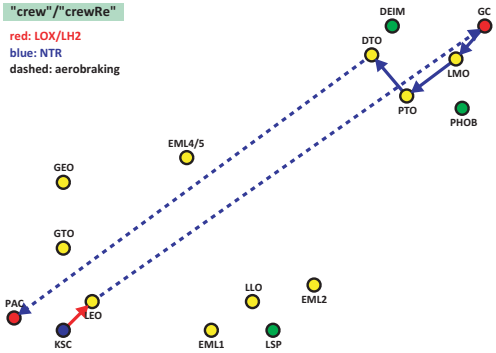
Downloaded by UNIVERSITY OF TENNESSEE on October 15, 2015 | http://arc.aiaa.org | DOI: 10.2514/6.2013-5414



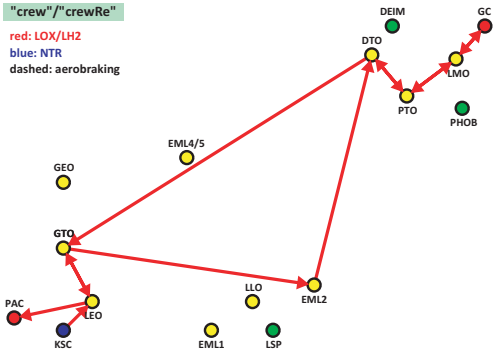
(a) Baseline scenario



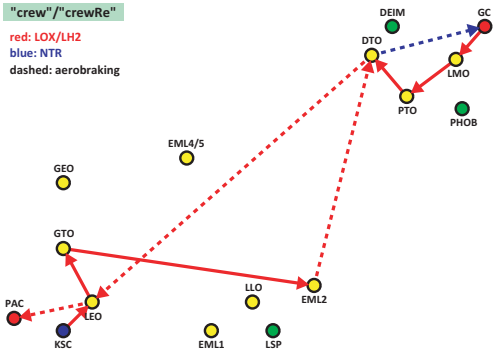
(b) (1) LOX/LH_2 only with aerobraking option



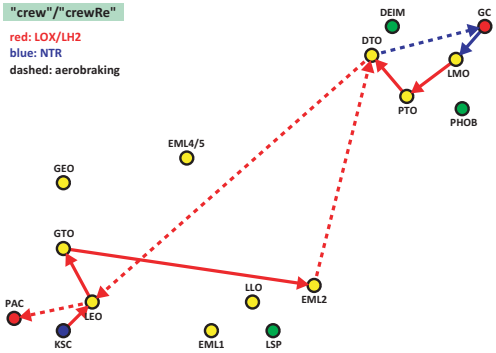
(c) (2) NTR only with aerobraking option



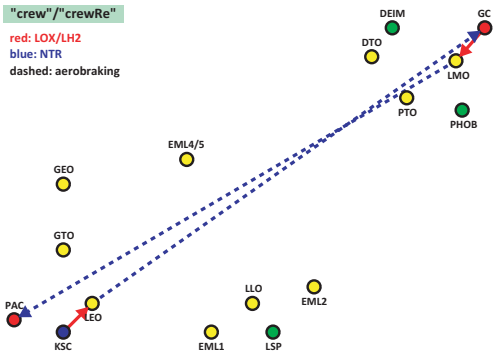
(d) (3) LOX/LH_2 and NTR without aerobraking option



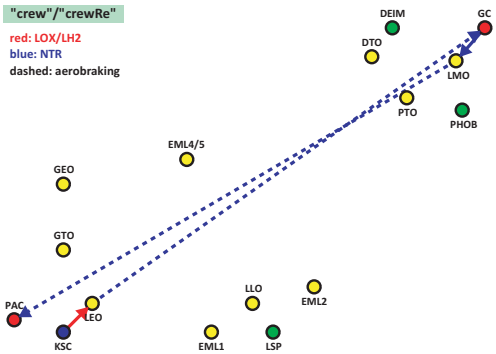
(e) (4) ISRU available on the Moon and Mars



(f) (5) ISRU available on the Moon only



(g) (6) ISRU available on Mars only



(h) (7) ISRU not available

Figure 14. Flows of "crew" and "crewRe"

References

[1]Obama, B., "Remarks by the President on Space Exploration in the 21st Century," John F. Kennedy Space Center, Merritt Island, FL.

[2]Zubrin, R., "Human Mars Exploration: The Time Is Now," *Journal of Cosmology*, Vol. 12, 2010, pp. 3549–3557.

[3]Ishimatsu, T., Grogan, P. T., and de Weck, O. L., "Interplanetary Trajectory Analysis and Logistical Consideration of Human Mars Exploration," *Journal of Cosmology*, Vol. 12, 2010, pp. 3588–3600.

[4]Grogan, P. T., *A Flexible, Modular Approach to Integrated Space Exploration Campaign Logistics Modeling, Simulation, and Analysis*, Master's Thesis, Massachusetts Institute of Technology, Cambridge, MA, September 2010.

[5]Ishimatsu, T., de Weck, O. L., Hoffman, J. A., and Ohkami, Y., "A Proposal for Graph-Theoretic Modeling Approach to Resource-Economy in Spaceflight Campaign Logistics," AIAA 2011-7347, AIAA SPACE 2011 Conference & Exposition, Long Beach, CA, September 2011.

[6]de Weck, O. L. and Simchi-Levi, D., "Haughton-Mars Project Expedition 2005," Final Report, NASA/TP 2006-214196.

[7]Shull, S. A., Gralla, E. L., de Weck, O. L., and Shishko, R., "The Future of Asset Management for Human Space Exploration: Supply Classification and an Integrated Database," AIAA 2006-7232, AIAA SPACE 2006 Conference & Exposition, San Jose, CA, September 2006.

[8]Taylor, C., *Integrated Transportation System Design Optimization*, PhD Thesis, Massachusetts Institute of Technology, Cambridge, MA, February 2007.

[9]Gralla, E. L., Shull, S. A., de Weck, O. L., Lee, G. Y., and Shishko, R., "A Modeling Framework for Interplanetary Supply Chains," AIAA 2006-7229, AIAA SPACE 2006 Conference & Exposition, San Jose, CA, September 2006.

[10]de Weck, O. L., Silver, M. R., and Shishko, R., "Measures of Effectiveness for SpaceNet," Unpublished, v3.3.

[11]Siddiqi, A., de Weck, O. L., Lee, G. Y., and Shull, S. A., "Matrix Modeling Methods for Spaceflight Campaign Logistics Analysis," *Journal of Spacecraft and Rockets*, Vol. 46(5), 2009, pp. 1037–1048.

[12]Grogan, P. T., Siddiqi, A., and de Weck, O. L., "Matrix Methods for Optimal Manifesting of Multi-Node Space Exploration Systems," *Journal of Spacecraft and Rockets*, Vol. 48(4), 2010, pp. 679–690.

[13]de Weck, O. L., Simchi-Levi, D., Shishko, R., Ahn, J., Gralla, E. L., Klabjan, D., Mellein, J., Shull, S. A., Siddiqi, A., Bairstow, B. K., and Lee, G. Y., "SpaceNet v1.3 User's Guide," NASA/TP 2007-214725.

[14]Lee, G. Y., Jordan, E., Shishko, R., de Weck, O. L., Armar, N., and Siddiqi, A., "SpaceNet: Modeling and Simulating Space Logistics," AIAA 2008-7747, AIAA SPACE 2008 Conference & Exposition, San Diego, CA, September 2008.

[15]Grogan, P. T., Siddiqi, A., and de Weck, O. L., "A Flexible Architecture and Object-oriented Model for Space Logistics Simulation," AIAA 2009-6548, AIAA SPACE 2009 Conference & Exposition, Pasadena, CA, September 2009.

[16]de Weck, O. L., Jordan, E. O., Lee, G. Y., Grogan, P. T., Ishimatsu, T., and Siddiqi, A., "SpaceNet 2.5 User's Guide," .

[17]Ishimatsu, T., de Weck, O. L., Hoffman, J. A., Ohkami, Y., and Shishko, R., "A Graph-Theoretic Modeling Framework for Resource-Economy in Space Logistics," AIAA 2012-5125, AIAA SPACE 2012 Conference & Exposition, Pasadena, CA, September 2012.

[18]Ishimatsu, T., *Generalized Multi-Commodity Network Flows: Case Studies in Space Logistics and Complex Infrastructure Systems*, PhD Thesis, Massachusetts Institute of Technology, Cambridge, MA, June 2013.

[19]Sanders, G. B., Larson, W. E., Sacksteder, K. R., and Mclemore, C. A., "NASA In-Situ Resource Utilization (ISRU) Project - Development & Implementation," AIAA 2008-7853, AIAA SPACE 2008 Conference & Exposition, San Diego, CA, September 2008.

[20]Chepko, A., *Technology Selection and Architecture Optimization of In-Situ Resource Utilization Systems*, Master's Thesis, Massachusetts Institute of Technology, Cambridge, MA, June 2009.

[21]Sanders, G. B., Carey, W. C., Piedboeuf, J.-C., and Lorenzoni, A., "Lunar In-Situ Resource Utilization in the ISECG Human Lunar Exploration Reference Architecture," IAC-10.A5.1.7, IAF 61st International Astronautical Congress, Prague, Czech, October 2010.

[22]Wittenberg, L. J., "In-Situ Extraction of Lunar Soil Volatiles," Tech. Rep. WC SAR-TR-AR3-9311-3, Wisconsin Center for Space Automation and Robotics, University of Wisconsin, November 1993.

[23]Cordell, B. M., "The Moons of Mars: A Source of Water for Lunar Bases and LEO," *Lunar Bases and Space Activities of the 21st Century*, Vol. 1, 1985, pp. 809–816.

[24]Muscatello, A. C., Mueller, R. P., Sanders, G. B., and Larson, W. E., "Phobos and Deimos Sample Collection and Prospecting Missions for Science and ISRU," *LPI Contributions*, Vol. 1679, No. 4296, 2012.

[25]Ahuja, R. K., Magnanti, T. L., and Orlin, J. B., *Network Flows – Theory, Algrithims, and Applications*, Prentice Hall, 1993.

[26]Drake, B. G., "Human Exploration of Mars: Challenges and Design Reference Architecture 5.0," *Journal of Cosmology*, Vol. 12, 2010, pp. 3578–3587.

[27]Drake, B. G., "Human Exploration of Mars Design Reference Architecture 5.0," NASA/SP 2009-566, NASA Johnson Space Center, Houston, TX.

[28]Zegler, F. and Kutter, B., "Evolving to a Depot-Based Space Transportation Architecture," AIAA 2010-8638, AIAA SPACE 2010 Conference & Exposition, Anaheim, CA, August-September 2010.

[29]Oeftering, R. C., "A Cis-Lunar Propellant Infrastructure for Flexible Path Exploration and Space Commerce," AIAA 2011-7113, AIAA SPACE 2011 Conference & Exposition, Long Beach, CA, September 2011.

[30]Ishimatsu, T., Hoffman, J. A., and de Weck, O. L., "Method for Rapid Interplanetary Trajectory Analysis using ΔV Maps with Flyby Options," *Journal of the British Interplanetary Society*, Vol. 64, 2011, pp. 204–213.

[31]Wikipedia, "Delta-v budget. URL: http://en.wikipedia.org/wiki/Delta-v_budget [cited April 3, 2013]," .

Downloaded by UNIVERSITY OF TENNESSEE on October 15, 2015 | http://arc.aiaa.org | DOI: 10.2514/6.2013-5414